Electronic Companion

EC.1 Statement Proofs

Proof of Proposition 1. We prove it by transforming the capacitated vehicle routing problem (CVRP), which is known to be $\mathcal{NP}$-hard, to the GDP.

The CVRP involves finding the least-cost routes for a fleet of vehicles to service a set of customers, subject to vehicle capacity constraints. In the following, we show that any instance of the CVRP can be transformed into an instance of the GDP in polynomial time.

We set the number of clusters L to be equal to the number of customer vertices in the CVRP instance. For each such vertex, we create a cluster $\mathcal{N}_l$ containing exactly one customer. Since each cluster has size 1, no cable-trench costs are included in the objective function. Additionally, we set $C_l = 0$ for each $\mathcal{N}_l \in \mathcal{N}$, leaving only the vehicle routing cost, which is the sum of the distances traveled by the vehicles. The only remaining constraints are the vehicle capacity constraints due to the generator weights.

Clearly, any feasible solution to this GDP instance corresponds to a feasible solution to the CVRP instance, with the same objective cost, and vice versa. This demonstrates that solving the GDP instance is at least as hard as solving the CVRP instance, thus completing the proof. □

Proof of Proposition 2. Consider a customer subset $\mathcal{C} \subseteq \mathcal{N}_l$, where $l \in \{1,\ldots,L\}$. Let $m \in \mathcal{C}$ denote the root vertex of the cable-trench network for this subset. From (EC.7b), the total downtime for the subset $\mathcal{C}$, denoted $\rho(\mathcal{C})$, is given by:

$$\rho(\mathcal{C}) = \rho_m + \sum_{i\in\mathcal{C}, i\neq m} \rho_i. \tag{EC.1}$$

Since $y_{i,m} = 1$ for all $i \in \mathcal{C}$, the downtime expression (EC.7b) simplifies to:

$$\rho_i = \rho_m + \rho_i', \quad \text{where} \quad \rho_i' = \sum_{(u,v)\in\mathcal{N}_l} (t^{\gamma}_{u,v} + t^{\tau}_{u,v}) g_{i,m,u,v} + s_i, \quad \forall i \in \mathcal{C}, i \neq m.$$

Substituting this expression into (EC.1), we have:

$$\begin{aligned}
\rho(\mathcal{C}) &= \rho_m + \sum_{i\in\mathcal{C}, i\neq m} (\rho_m + \rho_i') \\
&= \rho_m + \sum_{i\in\mathcal{C}, i\neq m} \rho_m + \sum_{i\in\mathcal{C}, i\neq m} \rho_i' \\
&= |\mathcal{C}|\rho_m + \sum_{i\in\mathcal{C}, i\neq m} \rho_i'.
\end{aligned}$$

From the final expression, the total downtime $\rho(\mathcal{C})$ can be decomposed into two components. The generator delivery time is given by $|\mathcal{C}|\rho_m$, which represents the propagation of downtime across the subset $\mathcal{C}$ due to the root vertex m. The value ρ_m, representing the generator delivery time, is computed from (EC.7a) and (EC.7c). The network construction time is given by $\sum_{i\in\mathcal{C}, i\neq m} \rho_i'$, which accounts for the downtime contributions from individual customers in $\mathcal{C}$ due to cabling and trenching. □

Proof of Proposition 3. We prove that the pricing problem P_T is at least as hard as the Minimum-Weight Rooted (Non-Necessarily Spanning) Arborescence Problem (MRAP) with unrestricted-in-sign weights on arcs through a polynomial-time reduction, where MRAP is shown to be $\mathcal{NP}$-hard in Rao and Sridharan (2002).

Let $\mathcal{G} = (\mathcal{V}, \mathcal{A})$ be a directed graph with the arc weights $\bar{u}_{i,j}$ associated with each arc $(i,j) \in \mathcal{A}$. Let $m' \in \mathcal{V}$ be the root node, which has no incoming arcs and a single outgoing arc $(m',m) \in \mathcal{A}$. An arborescence T is a connected subgraph of $\mathcal{G}$ with no cycles such that there is a directed path from m' to every other vertex in T. The cost of arborescence T is the sum of the weights of the arcs in T. The MRAP consists of finding a minimum-cost arborescence T in $\mathcal{G}$, where T includes the root arc (m',m), and no two arcs in T are directed toward the same node.

We use the method of restriction (Johnson and Garey 1979) by allowing only instances in which $E_l = |\mathcal{N}_l|$ and $C_l = 0$ for all $\mathcal{N}_l \in \mathcal{N}$, and having $\gamma_a = \gamma_c = \gamma_t = 0$. To reduce MRAP to the pricing problem P_T of the root vertex m in cluster $\mathcal{N}_l \in \mathcal{N}$, we proceed as follows. The weight of arc (m',m) in $\mathcal{A}$ is set to $\bar{u}_{m',m} =$

$\eta_m + \mu_m - \pi_m - \nu_l$, and the weight of arcs $(i,j) \in \mathscr{A} \setminus \{(m',m)\}$ is set to $\bar{u}_{i,j} = c^{\tau}_{i,j} - \pi_j + \eta_m$. Note that the arc weights $\bar{u}_{i,j}$ can have differing signs by definition of the dual variables in Section 4.3.1.

The transformation described above ensures that the optimal solution of MRAP in $\mathscr{G}$ has the same total cost as the optimal solution for the pricing problem P_T in $\mathscr{G}$. The converse is also true. Therefore, the P_T is $\mathcal{NP}$-hard. □

Proof of Proposition 4. We prove it by presenting a recursive DP algorithm that solves a relaxation of P_T in pseudopolynomial time.

Problem P_T is solved for each vertex m, which has a generator, in cluster $\mathcal{N}_l$ with $l \in \{1,\dots,L\}$. Define RP_T as a relaxation of problem P_T, allowing certain vertices and arcs to appear in the cable-trench network of m more than once. Let $\Psi(i,j,q)$ represent the minimum cost of constructing a cable-trench network rooted at vertex $i \in \mathcal{N}_l$, containing q vertices, such that every child $k \in \mathcal{N}_l$ of i satisfies $k \le j$, i.e., an outgoing arc from root i to child k exists only if the index of k is less than or equal to j. Recall that E_l is the size of the largest network that satisfies the generator capacity constraints in cluster $\mathcal{N}_l$. Let n_l be the largest customer index in cluster $\mathcal{N}_l$. The functions $\Psi(i,j,q)$ are computed for all $i,j \in \mathcal{N}_l$ and $q \in \{1,\dots,E_l\}$, using the recursive formulation below:

$$\Psi(i,j,q) = \min \begin{cases} \Psi(i,j-1,q), & \textit{(i)} \\ \bar{c}(i,j) + \Psi(j,n_l,q-1), & \textit{(ii)} \\ \min\limits_{1 \le q' \le q-1} \left\{ \Psi(i,j-1,q-q') + \bar{c}'(i,j) + \Psi(j,n_l,q') \right\}. & \textit{(iii)} \end{cases}$$

In the above recursion, case (i) represents the state in which root i has no outgoing arc to vertex j. Meanwhile, in case (ii), the subtree rooted at j of size $q-1$ is the unique child of i. In case (iii), the subtree of size q is obtained by connecting a subtree rooted at j of size q' to a subtree rooted at i of size $q-q'$. The edge costs $\bar{c}(i,j)$ and $\bar{c}'(i,j)$ in cases (ii) and (iii), respectively, are computed as $\bar{c}(i,j) = \eta_m - \pi_i + (q-1)c^{\gamma}_{i,j} + c^{\tau}_{i,j} + C_l((q-1)(t^{\gamma}_{i,j} + t^{\tau}_{i,j}) + s_i)$, and $\bar{c}'(i,j) = q'c^{\gamma}_{i,j} + c^{\tau}_{i,j} + C_l q'(t^{\gamma}_{i,j} + t^{\tau}_{i,j})$. The reduced cost of the resulting column that corresponds to vertex m in cluster $\mathcal{N}_l$ is given by $\min_{1 \le q \le E_l}\{\Psi(m,n_l,q)\}$. Since the complexity of case (iii) is $O(E_l)$ and there are $O(|\mathcal{N}_l|^2 E_l)$ subproblems, the overall complexity of computing $\Psi(i,j,q)$ for the relaxation RP_T is $O(|\mathcal{N}_l|^2 E_l^2)$. □

Proof of Proposition 5. Consider a subset of customer clusters S in $\mathscr{S}$. Define $\Omega(S) = \sum_{\mathcal{N}_l \in S} \sum_{i \in \mathcal{N}_l} \sum_{t \in \mathcal{T}(i)} \Omega_t \chi_t$ as the total weight of the trees in S. For a given feasible solution, the number of vehicles required to service all tree roots in S is at least $\lceil \Omega(S)/Q \rceil$. The total capacity of these vehicles is $Q\lceil \Omega(S)/Q \rceil$. We say that a solution violates the vehicle capacity constraint with respect to S if the total tree weight $\Omega(S)$ exceeds the capacity of the vehicles servicing S. Thus, the following inequalities hold:

$$Q \sum_{r \in \hat{\mathcal{R}}_S} \theta_r \ge Q \left\lceil \frac{\Omega(S)}{Q} \right\rceil \ge \Omega(S) = \sum_{\mathcal{N}_l \in S} \sum_{i \in \mathcal{N}_l} \sum_{t \in \mathcal{T}(i)} \Omega_t \chi_t,$$

which completes the proof. □

Proof of Proposition 6. Consider a subset of customer clusters $S \subseteq \mathscr{S}$. Let $\Omega_l = \sum_{i \in \mathcal{N}_l} \sum_{t \in \mathcal{T}(i)} \Omega_t \chi_t$, and $\Omega(S) = \sum_{\mathcal{N}_l \in S} \Omega_l$ denote the total weight of the trees in S. Each route can service at most Q units of weight, and the total capacity provided by the routes in $\hat{\mathcal{R}}_S$ is $Q \sum_{r \in \hat{\mathcal{R}}_S} \theta_r$. In an optimal solution, the number of routes required to serve S is at least $\lceil \Omega(S)/Q \rceil$, since each route has a capacity of Q. Hence, the following relation holds:

$$Q \sum_{r \in \hat{\mathcal{R}}_S} \theta_r \ge Q \left\lceil \frac{\Omega(S)}{Q} \right\rceil,$$

for all $S \subseteq \mathscr{S}$. Although $\Omega(S)$ is part of decisions, by definition of the bounding problem F_M in EC.4.3, which minimizes the total tree weight, we have $\bar{\Omega}_l \le \Omega_l$ for each cluster l, where $\bar{\Omega}_l$ is a valid lower bound on the weight of cluster l, as defined in Section 4.3.3. Therefore, the following relations hold:

$$\left\lceil \frac{\sum_{\mathcal{N}_l \in S} \Omega_l}{Q} \right\rceil \ge \sum_{r \in \hat{\mathcal{R}}_S} \theta_r \ge \left\lceil \frac{\sum_{\mathcal{N}_l \in S} \bar{\Omega}_l}{Q} \right\rceil,$$

which completes the proof. □

Proof of Proposition 7. Consider a pair of states, denoted as (I, I') with $I = (i, U, w, \Xi, Y)$ and $I' = (i, U', w', \Xi', Y')$. Let P denote the *completion* of state I, which is any path from vertex $j \in \mathcal{V} \setminus \{i\}$ that ends at depot o^-. Let $\mathscr{F}$ be the set of all feasible completions of state I. Completion P is feasible if $V(P) \cap U = \emptyset$, where $V(P)$ is the set of visited vertices; no cluster is entered more than once, and the number of delivered generators in each cluster $\mathcal{N}_l \in \mathcal{N}$ is less than or equal to p_l. Due to $U \subseteq U'$ and $w \leq w'$, for each completion $P \in \mathscr{F}'$, we have $P \in \mathscr{F}$. Let $\upsilon(P)$ and $\varkappa(P)$ denote the sets of binary resources for completion P associated with constraints (5) and (6), respectively. The following relations hold:

$$(\upsilon, \upsilon', \upsilon(P)) \in \left\{ \begin{matrix} (0,0,0), & (0,0,1), & (0,1,0), & (0,1,1), \\ (1,0,0), & (1,0,1), & (1,1,0), & (1,1,1) \end{matrix} \right\}.$$

The same definition applies to the binary resources associated with Y.

Define $c(P)$ as the reduced cost of completion P due to arc traversal, i.e., the cost in (4), vehicle's travel cost, and dual contributions due to constraints (3c). Let $(P \oplus I)$ (resp., $(P \oplus I')$) denote the complete path obtained by joining state I (resp., I') with completion P. Because the same completion P is appended to both states, the difference between the reduced cost of $(P \oplus I)$, denoted as $c(P \oplus I)$, and $(P \oplus I')$, denoted $c(P \oplus I')$, is calculated by:

$$\begin{aligned}
& c(P \oplus I) - c(P \oplus I') \\
& \quad = \Pi(i, U, w, \Xi, Y) + c(P) - \sum_{(\upsilon,\upsilon') \in \Xi \times \Xi' : \upsilon=0, \upsilon'=1, \upsilon(P)=1} \xi(\upsilon') - \sum_{(\upsilon,\upsilon') \in \Xi \times \Xi' : \upsilon=0, \upsilon'=0, \upsilon(P)=1} \xi(\upsilon') \\
& \quad - \sum_{(\varkappa,\varkappa') \in Y \times Y' : \varkappa=0, \varkappa'=1, \varkappa(P)=1} \xi(\varkappa') - \sum_{(\varkappa,\varkappa') \in Y \times Y' : \varkappa=0, \varkappa'=0, \varkappa(P)=1} \xi(\varkappa') \\
& \quad - \Pi'(i, U', w', \Xi', Y') - c(P) + \sum_{(\upsilon,\upsilon') \in \Xi \times \Xi' : \upsilon=1, \upsilon'=0, \upsilon(P)=1} \xi(\upsilon) + \sum_{(\upsilon,\upsilon') \in \Xi \times \Xi' : \upsilon=0, \upsilon'=0, \upsilon(P)=1} \xi(\upsilon') \\
& \quad + \sum_{(\varkappa,\varkappa') \in Y \times Y' : \varkappa=1, \varkappa'=0, \varkappa(P)=1} \xi(\varkappa) + \sum_{(\varkappa,\varkappa') \in Y \times Y' : \varkappa=0, \varkappa'=0, \varkappa(P)=1} \xi(\varkappa') \\
& \quad = \Pi(i, U, w, \Xi, Y) - \sum_{(\varkappa,\varkappa') \in Y \times Y' : \varkappa=0, \varkappa'=1, \varkappa(P)=1} \xi(\varkappa') - \sum_{(\upsilon,\upsilon') \in \Xi \times \Xi' : \upsilon=0, \upsilon'=1, \upsilon(P)=1} \xi(\upsilon') \\
& \quad - \Pi'(i, U', w', \Xi', Y') + \sum_{(\upsilon,\upsilon') \in \Xi \times \Xi' : \upsilon=1, \upsilon'=0, \upsilon(P)=1} \xi(\upsilon) + \sum_{(\varkappa,\varkappa') \in Y \times Y' : \varkappa=1, \varkappa'=0, \varkappa(P)=1} \xi(\varkappa).
\end{aligned} \tag{EC.2}$$

Because $\xi_S^{(5)} \geq 0$ and $\xi_S^{(6)} \geq 0$ for all $S \in \mathscr{S}$, from equation (EC.2), we obtain:

$$\begin{aligned}
c(P \oplus I) - c(P \oplus I') & \leq \Pi(i, U, w, \Xi, Y) - \Pi'(i, U', w', \Xi', Y') \\
& \quad + \sum_{(\upsilon,\upsilon') \in \Xi \times \Xi' : \upsilon=1, \upsilon'=0, \upsilon(P)=1} \xi(\upsilon) + \sum_{(\varkappa,\varkappa') \in Y \times Y' : \varkappa=1, \varkappa'=0, \varkappa(P)=1} \xi(\varkappa) \\
& \leq 0.
\end{aligned}$$

Therefore, every feasible completion P of state I' is also feasible for state I, and appending P to I yields a complete path with no larger reduced cost than appending it to I'. Thus, state I dominates state I'. □

Proof of Proposition 8. The left and right branching conditions (respectively, (8) and (9)) must ensure that the solution respects the vehicle capacity constraint. Consider an infeasible solution in which set $\mathcal{C} \subseteq \mathcal{V}$ is visited by a single vehicle, and its total weight, denoted as $\Omega(\mathcal{C})$, violates capacity Q. There are two possible scenarios regarding the optimal solution:

(i) All customer locations in $\mathcal{C}$ are visited and used as roots for trees in an optimal solution to the problem. In that case, vertex set $\mathcal{C}$ is either visited by multiple vehicles or by a single route. In the former case, CFIs (8) (the left-branch CFIs) enforce the minimal number of vehicles visiting this customer set without changing its total weight. In the latter case, trees rooted at $i \in \mathcal{C}$ must be reconstructed so that $\Omega(\mathcal{C}) \leq Q$ holds, as specified by (9) (the right-branch CFIs), allowing a single vehicle to service the set without violating the capacity.

(ii) Only a (possibly empty) subset of customer locations $\mathcal{C}' \subset \mathcal{C}$ is contained in an optimal solution as roots for trees. In this case, constraints (9) (the right-branch CFIs) are imposing the requirement regarding the vehicle capacity. □

Proof of Dominance 1. Define a *completion* P_u of the upward subtree $\mathcal{L}_u(i,q)$ as the set of a downward subtree T_0 and several upward subtrees $\{T_1,\ldots,T_j\}$, i.e., $P_u = \{T_0, T_1, \ldots, T_j\}$, which can be used to form a feasible complete tree $(\mathcal{L}_u(i,q), P_u)$ through concatenation with subtree $\mathcal{L}_u(i,q)$. Let $\mathscr{F}_u$ and $\mathscr{F}'_u$ be the sets of all feasible completions of subtrees $\mathcal{L}_u(i,q)$ and $\mathcal{L}'_u(i,q')$, respectively. For a subtree $T_g \in P_u$, let $c(T_g)$ be its reduced cost. Also, let $c(\mathcal{L}_u(i,q),P_u)$ be the reduced cost computed for the complete tree $(\mathcal{L}_u(i,q),P_u)$.

According to conditions $q \le q'$ and $\mathfrak{R}_{u,k} \le \mathfrak{R}'_{u,k}$ for all $k \in \mathcal{K}_u$, it is always feasible to concatenate subtrees $\{T_0, T_1, \ldots, T_j\} \in P_u$ with subtree $\mathcal{L}_u(i,q)$ for any completion $P_u \in \mathscr{F}'_u$, i.e., $\mathscr{F}'_u \subseteq \mathscr{F}_u$. The difference between the reduced costs of the complete tree $(\mathcal{L}_u(i,q),P_u)$ and $(\mathcal{L}'_u(i,q'),P_u)$ is:

$$c(\mathcal{L}_u(i,q),P_u) - c(\mathcal{L}'_u(i,q'),P_u) = \bar{c} + \sum_{g=0}^{j} c(T_g) - \bar{c}' - \sum_{g=0}^{j} c(T_g) = \bar{c} - \bar{c}'.$$

Therefore, we have:

$$c(\mathcal{L}_u(i,q),P_u) - c(\mathcal{L}'_u(i,q'),P_u) \le \bar{c} - \bar{c}' \le 0.$$

Thus, subtree $\mathcal{L}_u(i,q)$ has a more efficient resource consumption, and $\mathcal{L}'_u(i,q')$ can be discarded. □

Proof of Dominance 2. Define a *completion* P_d of the downward subtree $\mathcal{L}_d(i,q)$ as the set of upward subtrees $\{T_1,\ldots,T_j\}$, i.e., $P_d = \{T_1,\ldots,T_j\}$, which can be used to form a feasible complete tree $(\mathcal{L}_d(i,q),P_d)$ through concatenation with subtree $\mathcal{L}_d(i,q)$. Let $\mathscr{F}_d$ and $\mathscr{F}'_d$ be the sets of all feasible completions of subtrees $\mathcal{L}_d(i,q)$ and $\mathcal{L}'_d(i,q')$, respectively. For a subtree $T_g \in P_d$, let $c(T_g)$ be its reduced cost. Also, let $c(\mathcal{L}_d(i,q),P_d)$ be the reduced cost computed for the complete tree $(\mathcal{L}_d(i,q),P_d)$, and $P_d(q) \subseteq P_d$ be such a subset of upward subtrees that their sizes sum up to q.

According to conditions $q \le q'$ and $\mathfrak{R}_{d,k} \le \mathfrak{R}'_{d,k}$ for all $k \in \mathcal{K}_d$, and $\bar{a} + \nabla(q'') \le \bar{a}' + \nabla'(q'')$ for all $q'' \in \{1,\ldots,E_l - q'\}$, it is always feasible to concatenate subtrees $\{T_1,\ldots,T_j\} \in P_d$ with subtree $\mathcal{L}_d(i,q)$ for any completion $P_d \in \mathscr{F}'_d$, i.e., $\mathscr{F}'_d \subseteq \mathscr{F}_d$. The difference between the reduced costs of the complete tree $(\mathcal{L}_d(i,q),P_d)$ and $(\mathcal{L}'_d(i,q'),P_d)$ is computed for each q'' in $\{1,\ldots,E_l - q'\}$ as:

$$\begin{aligned} c(\mathcal{L}_d(i,q),P_d) - c(\mathcal{L}'_d(i,q'),P_d) = \bar{c} + &\\ &\sum_{g \in P_d(q'')} c(T_g) + \Upsilon(q'') - \bar{c}' - \sum_{g \in P_d(q'')} c(T_g) - \Upsilon'(q'') \\ &= \bar{c} + \Upsilon(q'') - \bar{c}' - \Upsilon'(q''). \end{aligned}$$

Therefore, we have:

$$c(\mathcal{L}_d(i,q),P_d) - c(\mathcal{L}'_d(i,q'),P_d) \le \bar{c} + \Upsilon(q'') - \bar{c}' - \Upsilon'(q'') \le 0, \quad \forall q'' \in \{1,\ldots,E_l - q'\},$$

and thus the subtree $\mathcal{L}'_d(i,q')$ can be safely discarded. □

EC.2 Compact Mixed Integer Linear Program Formulation

In the following, we present a compact MILP formulation of the GDP, referred to as F_1. The formulation respects the following set of operational constraints.

Define the arc-flow binary variables $x_{k,u,v}$, which are set to 1 if arc $(u,v) \in \mathcal{A}^1$ is traversed by vehicle k, and 0 otherwise. As each vehicle route must start and end at the respective depots, and a vehicle can enter a cluster at most once, the vehicle arc flow constraints are given by:

$$\sum_{v \in \mathcal{V}^-} x_{k,o^+,v} = \sum_{u \in \mathcal{V}^+} x_{k,u,o^-} = 1, \qquad \forall k \in \{1,\ldots,K\}, \tag{EC.3a}$$

$$\sum_{j \in \delta^+(i)} x_{k,i,j} - \sum_{j \in \delta^-(i)} x_{k,j,i} = 0, \qquad \forall i \in \mathcal{V}, k \in \{1,\ldots,K\}, \tag{EC.3b}$$

$$\sum_{i \in \mathcal{N}_l} \sum_{j \in \delta^+(\mathcal{N}_l)} x_{k,i,j} = \sum_{i \in \delta^-(\mathcal{N}_l)} \sum_{j \in \mathcal{N}_l} x_{k,i,j} \le 1, \qquad \forall \mathcal{N}_l \in \mathcal{N}, k \in \{1,\ldots,K\}. \tag{EC.3c}$$

To assign customers to generators, define the binary assignment variables $y_{i,j}$, which are set to 1 if customer i is connected to a generator at customer j, and 0 otherwise. We have:

$$\sum_{k=1}^{K} \sum_{j \in \delta^+(i)} x_{k,i,j} = y_{i,i}, \qquad \forall i \in \mathcal{V}, \tag{EC.4a}$$

$$\sum_{j \in \mathcal{N}(i)} y_{i,j} = 1, \qquad \forall \mathcal{N}_l \in \mathcal{N}, i \in \mathcal{N}_l, \tag{EC.4b}$$

$$\sum_{i \in \mathcal{N}_l} y_{i,i} = p_l, \qquad \forall \mathcal{N}_l \in \mathcal{N}. \tag{EC.4c}$$

Constraints (EC.4a) impose that a generator must be placed at a vertex that has been visited by a vehicle. Constraints (EC.4b) ensure that each customer is connected to a generator positioned in the cluster. Constraints (EC.4c) define the number of generators to be placed in each cluster.

The cable-trench network design is constrained using the binary variables $g_{i,j,u,v}$, which are equal to 1 if customer i in cluster $\mathcal{N}_l \in \mathcal{N}$ is connected to a generator at customer $j \in \mathcal{N}(i)$ and the path includes arc $(u,v) \in \mathcal{A}_l$, and the binary variables $b_{i,u,v}$, which are set to 1 if a generator is placed at customer i and trenching arc (u,v) is used to connect customer v to this generator, and 0 otherwise. We accordingly have:

$$\sum_{u \in \mathcal{N}_l} g_{i,j,v,u} - \sum_{u \in \mathcal{N}_l} g_{i,j,u,v} = \begin{cases} y_{i,j} \text{ if } v = j, \\ -y_{i,j} \text{ if } v = i, \\ 0 \quad \text{otherwise}, \end{cases} \qquad \forall v, i, j \in \mathcal{N}_l, i \neq j, \mathcal{N}_l \in \mathcal{N}, \tag{EC.5a}$$

$$g_{i,j,u,v} \leq b_{j,u,v}, \qquad \forall \mathcal{N}_l \in \mathcal{N}, i, j \in \mathcal{N}_l, (u,v) \in \mathcal{A}_l, \tag{EC.5b}$$

$$b_{i,u,v} \leq y_{i,i}, \qquad \forall \mathcal{N}_l \in \mathcal{N}, i \in \mathcal{N}_l, (u,v) \in \mathcal{A}_l, \tag{EC.5c}$$

$$y_{i,j} = \sum_{u \in \mathcal{N}(i)} b_{j,u,i}, \qquad \forall i \in \mathcal{V}, j \in \mathcal{N}(i), i \neq j. \tag{EC.5d}$$

Constraints (EC.5a) are the flow balance constraints on cabling. Constraints (EC.5b) allow the cable connections to be established only along the arcs furnished with trenching equipment. Constraints (EC.5c) stipulate that a trench path from a vertex can be constructed only if a generator has been placed at that vertex. A single trench path must lead to a customer connected to a generator as per (EC.5d).

Further, define the continuous variables $\Delta_{k,i}$, which compute the total material weight (cable, trench, and generator) needed to place a generator delivered by vehicle k to customer $i \in \mathcal{V}$. The following block is the capacity constraints:

$$\begin{aligned} \Delta_{k,j} \geq \sum_{i \in \mathcal{N}_l} \sum_{(u,v) \in \mathcal{A}_l} a^{\gamma}_{u,v} g_{i,j,u,v} + \sum_{(u,v) \in \mathcal{A}_l} a^{\tau}_{u,v} b_{j,u,v} + D y_{j,j} & \\ + Q\Big(\sum_{i \in \delta^-(j)} x_{k,i,j} - 1\Big), \qquad & \forall \mathcal{N}_l \in \mathcal{N}, j \in \mathcal{N}_l, k \in \{1, \ldots, K\}, \end{aligned} \tag{EC.6a}$$

$$\sum_{i \in \mathcal{V}} \Delta_{k,i} \leq Q, \qquad \forall k \in \{1, \ldots, K\}, \tag{EC.6b}$$

$$\sum_{i \in \mathcal{N}(j)} h_i y_{i,j} \leq W y_{j,j}, \qquad \forall j \in \mathcal{V}. \tag{EC.6c}$$

Constraints (EC.6a) are used to compute the vehicle loads. The vehicle capacity must be respected as per constraints (EC.6b). The generator capacity limitations are satisfied by constraints (EC.6c).

The variables related to the time-component are $\rho_i \geq 0$, and $w_{k,i} \geq 0$ for customer i and vehicle k. The former correspond to the total disruption time of customer i, while the latter are the service start times of vehicle k at customer i. These time components are constrained as:

$$\rho_i \geq \sum_{k=1}^{K} w_{k,i} + f_i + M(y_{i,i} - 1), \qquad \forall i \in \mathcal{V}, \tag{EC.7a}$$

$$\rho_i \geq \rho_j + \sum_{(u,v)\in\mathcal{A}_l} (t^{\gamma}_{u,v} + t^{\tau}_{u,v}) g_{i,j,u,v} + s_i + M'(y_{i,j} - 1), \qquad \forall \mathcal{N}_l \in \mathcal{N}, i, j \in \mathcal{N}_l, i \neq j, \tag{EC.7b}$$

$$w_{k,i} - w_{k,j} + M x_{k,i,j} \leq M - f_i - d_{i,j}, \qquad \forall (i,j) \in \mathcal{A}^1, k \in \{1,\ldots,K\}. \tag{EC.7c}$$

Constraints (EC.7a) are used to determine the time required to deliver and set up a generator if a vehicle visits the customer. The big-M value in these constraints is defined as $M = \lceil Q/D \rceil \max_{i\in\mathcal{V}^+, j\in\mathcal{V}}\{d_{i,j} + f_j\} + \max_{i\in\mathcal{V}}\{d_{i,o^-}\}$. Constraints (EC.7b) determine the time required to establish a connection to a placed generator by a cable. The value for M' is selected as $M' = M + \left(W/\min_{i\in\mathcal{V}}\{h_i\}\right)\left(\max_{(u,v)\in\mathcal{N}_l, \mathcal{N}_l\in\mathcal{N}}\{t^{\gamma}_{u,v} + t^{\tau}_{u,v}\} + \max_{i\in\mathcal{V}}\{s_i\}\right)$. Constraints (EC.7c) structure the time relationships between the consecutive visits of vertices. Variable domains are defined by (EC.8):

$$x_{k,u,v} \in \{0,1\},\ \forall (u,v) \in \mathcal{A}^1, k \in \{1,\ldots,K\}; \qquad y_{i,j} \in \{0,1\},\ \forall i \in \mathcal{V}, j \in \mathcal{N}(i), \tag{EC.8a}$$

$$b_{i,u,v} \in \{0,1\},\ \forall \mathcal{N}_l \in \mathcal{N}, i \in \mathcal{N}_l, (u,v) \in \mathcal{A}_l, \tag{EC.8b}$$

$$\Delta_{k,i} \geq 0,\ \forall i \in \mathcal{V}, k \in \{1,\ldots,K\}; \quad \rho_i \geq 0,\ \forall i \in \mathcal{V}; \qquad w_{k,i} \geq 0,\ \forall i \in \mathcal{V}, k \in \{1,\ldots,K\}. \tag{EC.8c}$$

In summary, F_1 is formulated as follows:

$$(F_1) \quad \min \underbrace{\sum_{\mathcal{N}_l\in\mathcal{N}} C_l \sum_{i\in\mathcal{N}_l} \rho_i}_{\text{Outage cost}} + \underbrace{\sum_{k=1}^{K} \sum_{(u,v)\in\mathcal{A}^1} e_{u,v} x_{k,u,v} + \sum_{\mathcal{N}_l\in\mathcal{N}} \sum_{i\in\mathcal{N}_l} \sum_{(u,v)\in\mathcal{A}_l} \left(c^{\tau}_{u,v} b_{i,u,v} + \sum_{j\in\mathcal{N}_l} c^{\gamma}_{u,v} g_{i,j,u,v} \right)}_{\text{Operational cost}} \tag{EC.9}$$
$$\text{s.t. (EC.3)} - \text{(EC.8)}.$$

Objective (EC.9) minimizes the total of the outage cost (based on the disruption time) and the operational cost (the routing cost and construction costs).

EC.3 Outline of the RR-BPC

Algorithm 1 outlines the RR-BPC approach.

Algorithm 1 The Relax-then-Repair-Driven Branch-Price-and-Cut Approach for the GDP

Step 1. *Relax*: Relax the MINLP F_2 by dropping constraints (3e) to obtain $\mathcal{F}$.

Step 2. *Initialization*: Compute an initial solution and an upper bound of the problem by executing procedure H_0, as described in Section EC.4.2.2. If H_0 terminates with a feasible solution, denoted as $\{X_0, \Theta_0\}$, of value z_0, set $UB = z_0$, and the best solution $\overline{X} = X_0$ and $\overline{\Theta} = \Theta_0$. Otherwise, set $UB = +\infty$. Use $\{X_0, \Theta_0\}$ to initialize sets $\hat{\mathcal{T}}$ and $\hat{\mathcal{R}}$.

Step 3. *Pricing*: Solve relaxation $\mathcal{F}_{lp}$. First, try the heuristics in Section EC.4.1. If no columns are found, follow the exact pricing procedure in Sections 4.3.1 and 4.3.2.

Step 4. *Cutting*: Tighten the lower bound of $\mathcal{F}_{lp}$ by separating inequalities MWI, CLI, GCI, and SRI, as described in Section 4.3.3. If a violated inequality is added to the model, return to step 3. Note that the GCIs and SRIs are only separated in nodes with a depth less than or equal to two.

Step 5. *Branch and Bound*:
  a. If $UB \leq LB$, prune the node.
  b. If the solution is fractional, go to step 5d.
  c. If it is integral but violates the vehicle capacity constraints, go to step 6. Otherwise, we have a feasible integer solution, and proceed as follows. Let z_i be the objective value at iteration i. If $z_i < UB$, update the upper bound by setting $UB = z_i$ and set the best solution to $\overline{X} = X_i$ and $\overline{\Theta} = \Theta_i$.
  d. Given a fractional solution $\{X_i, \Theta_i\}$ at the i-th iteration of the algorithm, go to step 5e if $\tilde{\chi}_t = 1$ for each $t \in X_i$. Otherwise, proceed to step 5f.
  e. Attempt to find an integer feasible solution by solving problem F_R (Section EC.4.2.1). If such a solution is found, update the upper bound if needed. Then, proceed to step 5f.
  f. Create two branch-and-bound nodes based on the branching rules described in Section 4.3.4. Sort the nodes according to the best bound criterion. For each node, perform step 3.
  g. If all the branch-and-bound nodes have been explored, output the optimal solution $\{\overline{X}, \overline{\Theta}\}$.

Step 6. *Repair*: If an integral solution violates the vehicle capacity constraints, invoke the repair procedure as described in Section 4.4.

EC.4 Enhancements

In this section, we discuss several enhancements to improve the tractability. First, we present procedures for solving pricing problems P_T and P_R heuristically in Section EC.4.1. Then, we describe two heuristics to obtain good upper bounds (feasible solutions) in Section EC.4.2. Based on a partial solution obtained during the BPC procedure, one procedure, called F_R, is proposed to find a feasible integer solution in Section EC.4.2.1. Then, we present a procedure to obtain an initial feasible solution for the GDP, called H_0, in Section EC.4.2.2.

EC.4.1 Heuristics for Pricing Problems

Solving the pricing problems P_T and P_R is the most time-consuming part of the proposed exact algorithm. To speed up this process, several heuristics are implemented. These heuristics are executed consecutively before running the exact algorithms discussed in Sections 4.3.1 and 4.3.2.

For the pricing problem P_T on tree variables, we have three heuristics. The first heuristic is adapted from Şahin and Yaman (2022). It limits the number of star-arbs that can be stored simultaneously in memory during the execution of the algorithm. Thus, it only considers more promising star-arbs. As a result, the algorithm requires many fewer steps to terminate, which, however, comes at the cost of possibly discarding an optimal column. The next heuristic is based on the arc-reduction technique in Fukasawa et al. (2006) and Şahin and Yaman (2022). More specifically, a vertex retains only a subset of incoming arcs according to some criterion, such as spatial proximity (Fukasawa et al. 2006) or dual values (Şahin and Yaman 2022). In practice, such reduced arc sets are much smaller than the complete set of arcs. The third heuristic relaxes a portion of the dominance rules, which is referred to as aggressive dominance in the literature (Desaulniers et al. 2008). Further, this heuristic, relaxing the dominance rules, is also applied to the pricing problem P_R.

EC.4.2 Heuristics for Upper Bounds

It is critical to find good-quality solutions early in the Branch-and-Bound process, which can help prune nodes and accelerate the search process. We discuss two procedures for that.

EC.4.2.1 Heuristic for Feasible Solutions Based on Partial Solutions.

During our extensive preliminary experiments, we observe that the solution often has integer values for the tree variables, but fractional values for the route variables. In this scenario, a feasible integer solution can be found in the following procedure. We first fix the tree variables to integer values in the current solution. Then the problem becomes one of minimizing vehicle routing costs. In our implementation, the latter problem, denoted as F_R, is solved by running a dynamic programming algorithm. The solution to F_R is subsequently used to update the upper bound of the problem if it has a better objective value. The details follow.

Here, we describe how procedure F_R is used to obtain an integer feasible solution, thus an upper bound, based on the solution in which the tree variables take integer values. Let $\mathcal{V}(X_k) \subseteq \mathcal{V}$ be the set of vertices corresponding to solution X_k at iteration k of the algorithm, such that each tree $t \in X_k$ is rooted at $i \in \mathcal{V}(X_k)$. Note that we have $\tilde{\chi}_t = 1$ for each $t \in X_k$. Let $\mathcal{R}^{\mathcal{F}}$ denote the set of vehicle routes defined for formulation F_R. The cost of route r in $\mathcal{R}^{\mathcal{F}}$ computes according to expression (1), where the service numbers $\Lambda_{i,r}$ are given by $\sum_{t\in\mathcal{T}(i)}\sum_{j\in\mathcal{N}(i)}\alpha_{j,t}\tilde{\chi}_t$ for all $i \in \mathcal{V}(X_k)$. For the sake of consistency, let θ_r denote binary variables that equal 1 if route $r \in \mathcal{R}^{\mathcal{F}}$ is selected in the solution, and 0 otherwise. Problem F_R is formulated as follows:

$$
\begin{aligned}
(F_R) \quad \min \ & \sum_{r\in\mathcal{R}^{\mathcal{F}}} c_r\theta_r && && \text{(EC.10)}\\
\text{s.t.} \ & \sum_{r\in\mathcal{R}^{\mathcal{F}}} \alpha_{i,r}\theta_r = 1, && \forall i \in \mathcal{V}(X_k), && \text{(EC.11)}\\
& \sum_{r\in\mathcal{R}^{\mathcal{F}}} \theta_r \le K, && && \text{(EC.12)}\\
& \theta_r \in \{0,1\}, && \forall r \in \mathcal{R}^{\mathcal{F}}. && \text{(EC.13)}
\end{aligned}
$$

Objective (EC.10) minimizes the total route cost. Constraints (EC.11) require that the corresponding tree roots be visited. The number of vehicles that must be used in the solution is constrained by (EC.12). Variable scopes are given by (EC.13).

The linear programming relaxation of F_R is solved by the algorithm described in Section 4.3.1. As only a subset of customer vertices is considered (i.e., $\mathcal{V}(X_k) \subseteq \mathcal{V}$), and the service numbers are provided by solution X_k, the resulting pricing problem is a simpler version of P_R. This makes solving F_R less computationally challenging. As the tree variables are obtained from the relaxed problem, F_R may fail to identify any feasible solutions, which may be due to the vehicle capacity constraints. Otherwise, the obtained solution provides an upper bound on $\mathcal{F}$.

EC.4.2.2 Heuristic for Initial Feasible Solutions.

To find an initial integer feasible solution to the problem, the following procedure, titled H_0, is implemented. First, the initial solution for the tree variables is obtained from the procedure for separating the MWIs, as described in EC.4.3. This partial solution guarantees that there exists a feasible solution for the route variables that does not violate the vehicle capacity constraints (otherwise, the problem is infeasible). Then, a feasible solution is obtained by solving for the route variables, as discussed in Section EC.4.2.1. The best integer solution found within the time limit, which is set to 60 seconds in our experiments, is used as the initial feasible solution. Moreover, its corresponding variables are used as the set of initial variables in the RR-BPC approach.

EC.4.3 Problem F_M for the Lower Bound $\bar{\Omega}_l$

The lower bound $\bar{\Omega}_l$ for each cluster $\mathcal{N}_l \in \mathcal{N}$ can be obtained by decomposing the problem by cluster and solving for the tree variables only. This can be achieved by solving problem F_M, which is detailed as follows. Let $\mathcal{T}^M$ denote the set of trees whose objective coefficients are the weights of the materials for constructing the corresponding cable-trench networks. As in formulation F_2, we use binary variables χ_t to denote whether tree $t \in \mathcal{T}^M$ is selected in the optimal solution. A feasible solution to F_M finds a partition of $\mathcal{V}$, visiting each customer exactly once, such that the number of trees constructed in each cluster $\mathcal{N}_l \in \mathcal{N}$ is exactly p_l. Its objective, denoted by $\bar{\Omega} = \sum_{\mathcal{N}_l \in \mathcal{N}} \bar{\Omega}_l$, minimizes the sum of the tree weights. To generate the set of the MWIs, F_M is formulated as:

$$(F_M) \quad \bar{\Omega} = \min \sum_{t \in \mathcal{T}^M} \Omega_t \chi_t \tag{EC.14}$$

$$\text{s.t.} \sum_{t \in \mathcal{T}^M} \alpha_{i,t} \chi_t = 1, \qquad \forall i \in \mathcal{V}, \tag{EC.15}$$

$$\sum_{i \in \mathcal{N}_l} \sum_{t \in \mathcal{T}^M(i)} \chi_t = p_l, \qquad \forall \mathcal{N}_l \in \mathcal{N}, \tag{EC.16}$$

$$\chi_t \in \{0,1\}, \qquad \forall t \in \mathcal{T}^M. \tag{EC.17}$$

Objective (EC.14) minimizes the total tree weight. Constraints (EC.15) impose that each customer must be connected to a tree, and the number of trees in each cluster $\mathcal{N}_l$ must be p_l for all $l \in \{1, \ldots, L\}$, as per constraints EC.16. The variable domains are given by (EC.17). To obtain lower bounds, the linear programming relaxation of F_M is solved using the algorithm introduced in Section 4.3.1.

EC.5 Computational Analyses

This section first evaluates the impact of the components in the RR-BPC approach in EC.5.1. This is followed by an analysis of the proposed star-arb relaxation in EC.5.2 and the bidirectional DP for tree generation in EC.5.3.

EC.5.1 Impact of the Components

We consider five different settings with BIS Instance-Set-50, which has 45 instances with 50 customers. We begin with the cases 'Base' and 'Base+E'. The former incorporates only the basic components outlined in Section 4.3. The latter configuration additionally utilizes the enhancements in Section EC.4. The third setting builds upon the 'Base+E' one by adding the inequalities GCI and SRI. The fourth setting incorporates the inequalities CLI and MWI into the 'Base+E' case. The last setting, 'All', integrates all components.

Figure EC.1 shows the number of instances solved optimally and the average CPU time achieved by each of the five settings in Instance-Set-50. The results show that the basic configurations 'Base' and 'Base+E'

Figure EC.1 The Impact of Components on Running Times and Number of Solved Instances in Instance-Set-50.

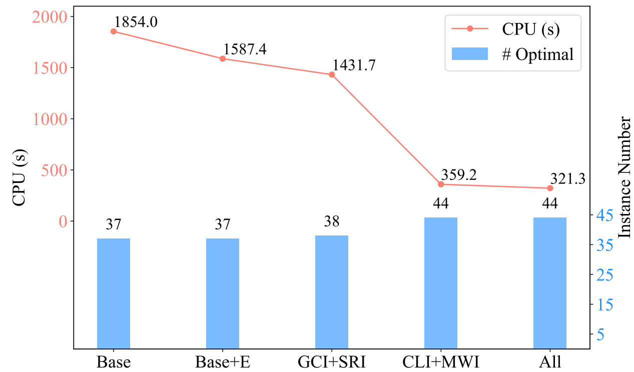


Table EC.1 Root Node Average Gaps (%) and Running Times (s).

		relax	q-arb	star-arb	GCI+SRI	CLI+MWI	All
50	gap	3.94	1.14	0.45	0.29	0.30	0.14
	cpu	1.12	0.88	0.81	1.61	82.73	93.71
75	gap	8.84	4.00	1.22	0.24	1.13	0.15
	cpu	3.60	5.93	6.37	9.82	7.04	10.94
100	gap	12.24	5.93	1.58	0.30	1.47	0.19
	cpu	9.83	12.86	15.04	19.99	15.16	23.16
Overall	gap	8.34	3.69	1.08	0.27	0.97	0.16
	cpu	4.85	6.56	7.40	10.48	34.98	42.60

solve the fewest instances within the TL, achieving optimality for 37 out of 45 instances. However, adding the enhancements enables the configuration ‘Base+E’ to achieve a 14.38% runtime improvement, reducing the runtime from 1,854.0 seconds to 1,587.4 seconds. Including the GCI and SRI inequalities yields a larger number of optimally solved instances (38/45) and a shorter runtime. Specifically designed for the problem at hand, the CLI and MWI inequalities have a greater impact, solving six more instances to optimality and yielding a fourfold speedup. When all components are applied under the ‘All’ setting, it optimally solves 44 out of 45 instances, with an average runtime of 321.3 seconds, which is over five times faster than the ‘Base’ case.

EC.5.2 Impact of the Relaxations

Next, we analyze the performance of the proposed star-arb relaxation for cable-trench networks (Section 4.3.1). Recall that star-arbs allow for the generation of non-elementary tree variables in the pricing problem P_T, with the exception of direct-path cycles of a length less than or equal to two, as well as parallel arcs. This relaxation is first compared against a purely non-elementary approach, which imposes no restrictions on the generated cycles, titled ‘*relax*’. Then, the star-arbs are compared against the q-arb relaxation of Uchoa et al. (2008), which forbids parallel arcs.

Table EC.1 presents the root node average gaps (in %) and CPU running times (in seconds) for different methods in Instance-Sets 50, 75, and 100, as well as overall averages. Although it has the lowest CPU times across all problem sizes, the non-elementary approach ‘*relax*’ has the highest average gap, increasing from 3.94% for 50 customers to 12.24% for 100 customers, resulting in an overall gap of 8.34%. The q-arb method improves the gap, with average values ranging from 1.14% for 50 customers to 5.93% for 100 customers, and an overall gap of 3.69%. The CPU times are moderately higher, averaging 6.56 seconds. The star-arb method achieves the best performance among these three methods in terms of solution quality, yielding the tightest bounds, with gaps consistently below 2% and an overall gap of 1.08%. This improvement comes at the cost of slightly higher CPU times than the q-arb, but the increase is marginal, suggesting that the star-arb strikes a strong balance between solution quality and overall algorithm performance.

Furthermore, Table EC.1 also shows the impact of the valid inequalities. Including ‘GCI+SRI’ and ‘CLI+MWI’ into the star-arb relaxation, the root node gaps are further reduced. Specifically, the average gaps drop from 1.08% for the star-arb method alone to just 0.27% and 0.97%, respectively. The combination of all cuts (‘All’) achieves the lowest overall gap of 0.16%, representing a significant improvement in solution quality. In summary, the combination of star-arb relaxation and valid inequalities reduces the root node optimality gap from 8.34% (the ‘*relax*’ approach) to 0.16%.

EC.5.3 Impact of Bidirectional DP for Tree Generation

To evaluate the computational contribution of the proposed bidirectional DP, we conduct an additional experiment comparing it with the mono-directional implementation that uses only the upward DP for tree generation. The results show that, without cutting planes, the average root-node solution time decreases from 7.10 seconds under the mono-directional DP to 5.93 seconds under the bidirectional DP. When cutting planes are added, the average solution time is reduced from 57.61 seconds to 50.47 seconds. Overall, the bidirectional DP yields an average computational improvement of 12.39%. Its benefit becomes more pronounced as the problem size increases, with average improvements of 14.56% and 22.81% for instances with 75 and 100 customers, respectively. We conclude that the bidirectional DP accelerates the tree-generation process, particularly for larger and more challenging instances.

EC. 6-14

In addition to this electronic companion, we provide a full version with extra details and supplementary numerical results. The full version EC is available on this link: `https://github.com/stackpilotphd/JOC-2025-12-OA-1872`.

EC.6 Notation

Table EC.2 Notation Summary

Symbol	Description
Sets	
$\mathcal{A}$	Set of all arcs
$\mathcal{A}^1$	Set of vehicle arcs
$\mathcal{A}^2$	Set of cable-trench arcs
$\mathcal{A}_l$	Set of arcs constituting cluster $\mathcal{N}_l$
$\mathcal{C}$	Subset of customers
$\delta^+(i)$	Set of succeeding vertices of customer i
$\delta^-(i)$	Set of preceding vertices of customer i
$\delta^+(\mathcal{N}_l)$	Set of succeeding vertices of cluster $\mathcal{N}_l$
$\delta^-(\mathcal{N}_l)$	Set of preceding vertices of cluster $\mathcal{N}_l$
$\mathcal{N}$	Set of clusters
$\mathcal{N}_l$	Set of customers constituting cluster l
$\mathcal{N}(i)$	Set of customers belonging to the same cluster as vertex i
$\mathcal{Q}_i$	Set of feasible service quantities at customer i
$\mathcal{R}$	Set of vehicle routes
$\mathcal{S}$	Non-empty subset of clusters
$\mathcal{T}$	Set of tree networks
$\mathcal{T}(i)$	Set of trees rooted at i
$\mathcal{V}$	Set of customer vertices
$\mathcal{V}^+$	Set of customer vertices and the start depot
$\mathcal{V}^-$	Set of customer vertices and the end depot
$\{o^+, o^-\}$	Set of start and end depots
Parameters	
$a^{\gamma}_{u,v}$	Cabling weight consumption over arc (u,v)
$a^{\tau}_{u,v}$	Trenching weight consumption over arc (u,v)
C_l	Outage cost of cluster $\mathcal{N}_l$
$c^{\gamma}_{u,v}$	Cabling cost over arc (u,v)
$c^{\tau}_{u,v}$	Trenching cost over arc (u,v)
$c_{u,v}$	Cost coefficient associated with arc (u,v)
D	Weight of a generator
$d_{u,v}$	Travel time of a vehicle over arc (u,v)
$e_{u,v}$	Travel cost of a vehicle over arc (u,v)
E_l	Maximal tree size in cluster $\mathcal{N}_l$
f_i	Vehicle service time at customer i
h_i	Electricity demand of customer i
K	Number of available vehicles
L	Number of clusters
$\vartheta_{u,v}$	Workers' walking distance over arc (u,v)
$\Lambda_{i,r}$	Service quantity at customer i in route r
p_l	Number of generators required by cluster $\mathcal{N}_l$
Q	Capacity of a vehicle
s_i	Time required to connect customer i by a cable
$t^{\gamma}_{u,v}$	Cabling time consumption over arc (u,v)
$t^{\tau}_{u,v}$	Trenching time consumption over arc (u,v)
W	Capacity of a generator
Ω_t	Weight of tree t
$\bar{\Omega}_l$	Weight lower bound of the trees in cluster $\mathcal{N}_l$
Variables	
$b_{i,u,v}$	Binary variables that take the value of 1 if a generator is at vertex i and trenching arc (u,v) connects customer v, and 0 otherwise
$g_{i,j,u,v}$	Binary variables that take the value of 1 if customer i is connected to generator j and the path includes arc (u,v), and 0 otherwise
$x_{k,u,v}$	Binary variables that take the value of 1 if arc $(u,v) \in \mathcal{A}^1$ is traversed by vehicle k, and 0 otherwise
$y_{i,j}$	Binary variables that take the value of 1 if customer i is connected to generator j, and 0 otherwise
$\Delta_{k,i}$	Continuous nonnegative variables that determine the weight of cabling and trenching materials delivered by vehicle k to customer i
ρ_i	Continuous nonnegative variables that determine the downtime of customer i
$w_{k,i}$	Continuous nonnegative variables that determine the service start time at customer i by vehicle k
θ_r	Binary variables that take the value of 1 if route r is selected, and 0 otherwise
χ_t	Binary variables that take the value of 1 if tree network t is selected, and 0 otherwise

EC.7 DP Algorithm for Tree Generation

First, the pseudocode for the upward tree generation is provided in Algorithm 2. Next, the complete DP procedure is outlined in Algorithm 3.

Algorithm 2 Upward DP for Tree Variables.

Input: Cluster index $l \in \{1,\dots,L\}$, root node index $m \in \mathcal{N}_l$.
Define: Upward subtree bucket set $\mathcal{U} = \bigcup_{q=1}^{\lfloor E_l/2 \rfloor} \mathcal{U}_q$.

for $i \in \mathcal{N}_l$ **do**
  Generate initial subtree $\mathcal{L}_u(i,1)$ and add it to $\mathcal{U}_1$.
end for
for $q = 2,\dots,\lfloor E_l/2 \rfloor$ **do**
  for $\mathcal{L}_u(i,q-1) \in \mathcal{U}_{q-1}$ **do**
    for $j \in \mathcal{N}_l$ **do**
      Perform Extension Case 1.
      if new label is not dominated **then**
        add to $\mathcal{U}_q$.
      end if
    end for
  end for
  for $q' = 1,\dots,q-1$ **do**
    for $\mathcal{L}_u(i,q-q') \in \mathcal{U}_{q-q'}$ **do**
      for $\mathcal{L}_u(j,q') \in \mathcal{U}_{q'}$ **do**
        Perform Extension Case 2.
        if new label is not dominated **then**
          add to $\mathcal{U}_q$.
        end if
      end for
    end for
  end for
end for
Return: $\mathcal{U}$.

Algorithm 3 Downward DP for Tree Variables.

Input: Cluster index $l \in \{1,\dots,L\}$, root node index $m \in \mathcal{N}_l$.
Define: Downward subtree bucket set $\mathcal{B} = \bigcup_{q=1}^{E_l} \mathcal{B}_q$.

Generate initial subtree $\mathcal{L}_d(m,1)$ and add it to $\mathcal{B}_1$.
Run Algorithm 2 to obtain set $\mathcal{U}$.
for $q = 2,\dots,E_l$ **do**
  for $q' = 1,\dots,q-1$ **do**
    for $\mathcal{L}_d(i,q-q') \in \mathcal{B}_{q-q'}$ **do**
      for $\mathcal{L}_u(j,q') \in \mathcal{U}_{q'}$ **do**
        Perform Extension Cases 1 and 2.
        if new labels are not dominated **then**
          add to $\mathcal{B}_q$.
        end if
      end for
    end for
  end for
end for
Return: minimum reduced cost tree from $\mathcal{B}$.

EC.8 Separation Algorithm for Inequalities (5)−(6)

The size and number of cluster subsets $S \in \mathscr{S}$ increases exponentially with the number of customer clusters, making it impractical to add all inequalities (5) and (6) directly to the model. Fortunately, the violated inequalities can be added ad hoc. In particular, a naive separation procedure enumerates all cluster subsets $S \in \mathscr{S}$ and checks whether the current solution at the end of column generation violates the CLIs or MWIs. Let $\theta(S)$ (resp., $\theta(\mathcal{N}_l)$) denote the subset of route variables entering set $S \in \mathscr{S}$ (resp., cluster $\mathcal{N}_l \in \mathcal{N}$) at least once in the current solution, and $\tilde{\theta}(S)$ denote the total number of such routes based on the values $\tilde{\theta}_r$ for all r in $\hat{\mathcal{R}}$ such that $\tilde{\theta}_r > 0$. To detect a violation, denoted by ϖ, the algorithm iterates over all clusters in set S and computes the value $\tilde{\theta}(S)$. This is outlined in Algorithm 4. If a violated inequality is identified, the column generation process is restarted.

Algorithm 4 Violation Check

procedure VIOLATION(S)
  $\theta(S) \leftarrow \emptyset$
  $\varpi \leftarrow 0$, $v \leftarrow 0$
  for $l \leftarrow 1$ **to** L **do**
    if $\mathcal{N}_l \in S$ **then**
      $\theta(S) \leftarrow \theta(S) \cup \theta(\mathcal{N}_l)$
    end if
  end for
  $v \leftarrow \tilde{\theta}(S)$
  $\varpi \leftarrow \begin{cases} 1/Q \sum_{\mathcal{N}_l \in S} \sum_{i \in \mathcal{N}_l} \sum_{t \in \hat{\mathcal{T}}(i)} \Omega_t \tilde{\chi}_t - v \text{ if CLI (5),} \\ \left\lceil \sum_{\mathcal{N}_l \in S} \bar{\Omega}_l / Q \right\rceil - v \text{ if MWI (6).} \end{cases}$
  return $\max\{\varpi, 0\}$
end procedure

Algorithm 5 Separation Routine

procedure SEPARATE
  $\mathscr{C} \leftarrow \emptyset$
  for $l \leftarrow 1$ **to** L **do**
    $S \leftarrow \{\mathcal{N}_l\}$
    while $|S| < MaxSize$ **do**
      $S \leftarrow \{\arg\max_{\mathcal{N}_{l'} \in \mathcal{N} \setminus S} \{\text{VIOLATION}(S \cup \mathcal{N}_{l'})\}\}$
    end while
    if VIOLATION$(S) > threshold$ **then**
      $\mathscr{C} \leftarrow \mathscr{C} \cup \{S\}$
    end if
  end for
  while $|\mathscr{C}| > MaxCut$ **do**
    $\mathscr{C} \leftarrow \mathscr{C} \setminus \arg\min_{S \in \mathscr{C}} \{\text{VIOLATION}(S)\}$
  end while
  return $\mathscr{C}$
end procedure

Table EC.3 Input Parameters for the GDP Instances.

Category	Parameter	Value	Unit	Source
Customer	Load	[10,25]	kW	Kansal and Tiwari (2024)
Vehicle	Capacity	20	ton	Zhao et al. (2021)
	Travel cost	3.13	USD/mile	UberFreight (2024)
Generator	Capacity	100	kW	Liu et al. (2021)
	Weight	2,976	kg	Cummins (2024)
Cable	Cost	3.94	USD/m	Luo et al. (2021)
	Weight	1.27	kg/m	Phelps Dodge International (2024)

When the number of clusters is large (typically with $L > 20$), the exact separation approach may be overly time-consuming. Therefore, for the sake of completeness, we provide a heuristic separation routine that can be invoked to identify violated inequalities (5) and (6). Algorithm 5 presents the separation routine. Let $\mathscr{C}$ be the set of violated inequalities separated in each round. The constants *MaxSize*, $threshold$ and *MaxCut* are user-defined parameters. The first one, *MaxSize*, denotes the maximum size of set S, which is set to L in our implementation. The next parameter, $threshold$, specifies the minimum violation value returned by Algorithm 4 that is required for an inequality to be added to the linear program. In our experiments, we set it to 1×10^{-6}. Lastly, the parameter *MaxCut* controls the number of violated inequalities added to the model at each run of the separation procedure, which we set to 100. The algorithm is briefly summarized as follows: Starting with a single cluster in set $\mathscr{C}$, the cluster that maximizes the violation is greedily added to the set until its size equals $MaxSize$. If the size of $\mathscr{C}$ exceeds the predefined parameter value, the procedure removes such a subset S of $\mathscr{C}$ that has the smallest violation.

EC.9 Instance Details

Instances for the GDP are based on open-source data on the National Shelter System Facilities designated by the Federal Emergency Management Agency (FEMA) (Homeland Security 2024). This database, designed for emergency management purposes across the United States, contains the geographical coordinates of emergency shelters – schools, churches, and community centers. We select five states in which hurricane occurrence is the most frequent (Statista 2024), namely, Florida (FL), Louisiana (LA), North Carolina (NA), South Carolina (SA), and Texas (TX). The instances are generated by the following steps.

We first determine the number of customers (set $\mathcal{V}$), which takes a value in $\{50, 75, 100\}$. Then, to determine the locations for each state, we use the geographical coordinates of shelters as cluster centers from which synthetic customer locations are generated. Specifically, we randomly sample L shelters from the FEMA database via Fischetti et al. (1997)'s clustering algorithm. Subsequently, random customer locations $i \in \mathcal{N}_l$ are generated for each cluster $\mathcal{N}_l$ ($l \in \{1, \ldots, L\}$) using VeRoViz, an open-source package that obtains real-world routing data from data providers (Peng and Murray 2022). Moreover, between these locations, VeRoViz is employed to generate realistic travel distances and times for both trucks and workers, which are used to create the arc set $\mathcal{A}$. In particular, for the vehicles, the arcs in $\mathcal{A}^1$ represent vehicle travel along the publicly accessible street network. For workers, the arcs in $\mathcal{A}^2$ represent foot movements along the pedestrian network (e.g., footpaths, sidewalks). Further, the generator set-up time f_i (including material unloading) is uniformly set to 20 minutes for each i in $\mathcal{V}$ in the baseline instances. These homogeneous set-up times reflect the clustered structure of the customer locations, where customers within the same spatial cluster face similar local conditions. In addition, we construct a second set of instances, which are used in the analyses in Section 5.4, such that values f_i vary randomly between 15 minutes and one hour. This variation captures practical differences, such as the need to locate a suitable vehicle-parking or generator-placement area. When a customer node is not directly accessible by the vehicle, values f_i also reflect the extra effort required for workers to carry materials and the generator from the nearest reachable point. Moreover, to model different scales of outages, we vary the number of clusters L in the set $\{10, 12, 14, 16, 20\}$. A total of three instances are generated for each combination of customer and cluster numbers, resulting in 45 base instances. An example of an instance with 10 clusters generated using VeRoViz is shown on the right in Figure EC.2.

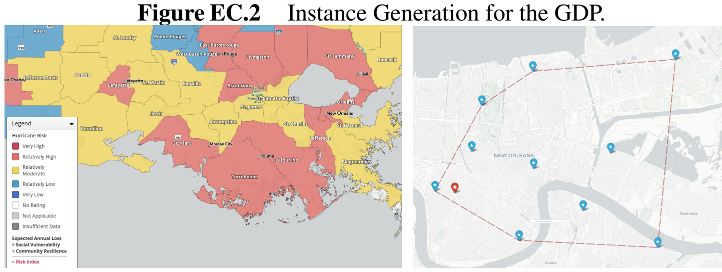
Further, the outage cost of cluster $\mathcal{N}_l$ is modeled as $C_l = \hat{C}_l \omega_l$ for $l \in \{1, \ldots, L\}$, where $\hat{C}_l$ is the disruption penalty per unit time, and ω_l is the assigned priority index. The disruption penalty per second is calculated as $\hat{C}_l = 0.5 \sum_{i \in \mathcal{N}_l} h_i / 3600$, where the cost of failing to satisfy the customer demand is set to 0.5 USD/kWh, as in Wan et al. (2025). Following Wang et al. (2022), randomly assigned indices ω_l are selected from the range $[1, 5]$, where 5 denotes the highest priority. Meanwhile, fleet parameter K in each instance is set to the minimum number of vehicles required to serve all clusters without violating the vehicle capacity constraints. We assume that the work time s_i required to connect customer i via a cable is a random variable with a uniform distribution over the interval [15, 30] minutes. For conciseness, the remaining parameters are listed in Table EC.3.

Taking into account that in practice, the overhead costs, weight, and time for trenching might vary (Vasko et al. 2002), for the basic instances we consider values of τ_k/γ_k in $\{0.5, 1.0, 1.5\}$ to construct the trench matrices, where the cabling coefficient γ_k is 1, for all $k \in \{a, c, t\}$. Therefore, three scenarios are considered by varying the cable-trench ratios for the 45 base instances, yielding a testbed of 135 instances, which are available through the link: `https://github.com/stackpilotphd/JOC-2025-12-OA-1872`. Furthermore, we generate an additional instance set to examine the impact of heterogeneous cable-trench parameter values. Specifically, each parameter in $\{\gamma_a, \gamma_c, \gamma_t, \tau_a, \tau_c, \tau_t\}$ is independently sampled from a uniform distribution over the interval $[0.5, 1.5]$. This yields a total of 45 instances. In summary, the computational experiments are conducted on three instance sets: the baseline instance set (BIS), the variable-service-time instance set (VST), and the variable-trench-coefficient instance set (VTC). All detailed results are provided in EC.13.2.

EC.10 Results of Solving MILP F_1.

This section presents the results of solving formulation F_1 with CPLEX on Instance-Set-50, which has a total of 45 instances with 50 customers. Key findings are summarized below in Table EC.4, with additional details available upon request.

Figure EC.2 Instance Generation for the GDP.



Note. Left: Counties in the state of LA which are prone to hurricane hits according to FEMA (2024). Right: An example of instance *orleans75*. Blue flags indicate shelters, and the red icon is the depot.

Table EC.4 Comparison of LP Relaxation, IP, and MIP Optimality Gaps (%) between F_1 (CPLEX) and F_2 (RR-BPC) in Instance-Set-50.

	MIP Gap				LP Relaxation				IP Gap			
	Overall	$\tau/\gamma = 0.5$	1.0	1.5	Overall	$\tau/\gamma = 0.5$	1.0	1.5	Overall	$\tau/\gamma = 0.5$	1.0	1.5
Max	46.45	46.45	40.95	37.87	55.47	55.47	50.25	46.73	28.04	28.04	5.14	4.86
Avg	28.19	33.15	27.76	23.99	36.38	41.81	35.86	31.48	2.20	3.37	1.64	1.60
Min	15.75	22.86	19.96	15.75	21.75	31.37	26.76	21.75	0.00	0.00	0.00	0.00

The analysis includes three metrics. The first metric, the MIP gap, represents the gap between the LP relaxation value and the best feasible solution at termination. We can see that applying F_1 to CPLEX cannot solve any of the tested instances to optimality within the specified TL. The second metric, LP relaxation, evaluates the quality of the bound at the root node of the solver's branch-and-bound tree relative to the optimal solution or the best known solution for unsolved instances. The LP relaxation of F_1 averages 36.38% of the optimal objective value. The minimum and maximum percentages are 21.75% and 55.47%, respectively. This suggests that the formulation is weak, which decreases the solver's efficiency. The third metric, the best feasible solution (IP gap), calculates the gap between the best integer solution found by CPLEX (Z_{F_1}) and the solution obtained by RR-BPC (Z_{F_2}), expressed as $(Z_{F_1}/Z_{F_2} - 1) \times 100\%$. The average optimality gap is 2.20%, with values ranging from 0.00% (indicating that the solution matches RR-BPC's optimal result) to 28.04%.

The data are further analyzed by grouping instances based on the cable-trench ratio ($\tau/\gamma = 0.5, 1.0, 1.5$, dropping subscripts a, c, and t). Instances with $\tau/\gamma = 0.5$ are notably more challenging for applying F_1 to CPLEX. A smaller τ/γ leads to a smaller material weight. While a smaller ratio decreases the number of vehicles required to serve all clusters, it lengthens the routes. Such characteristics make the problem more difficult.

EC.11 Robustness Analysis: Detailed Results and Implementation.

Table EC.5 Robust Analysis for BIS Instances with 50 Customers (%).

		$\varepsilon = 0.5$			$\varepsilon = 1.5$			$\varepsilon = 2.5$		
τ/γ	Γ	PoR	FRG	FRL	PoR	FRG	FRL	PoR	FRG	FRL
0.5	1	8.64	8.69	0.05	25.25	26.07	0.82	41.36	43.45	2.09
	3	9.93	9.96	0.03	29.43	29.89	0.46	48.65	49.81	1.16
	5	10.00	10.02	0.02	29.63	30.06	0.43	49.05	50.10	1.05
1.0	1	7.52	7.57	0.05	22.01	22.72	0.71	36.02	37.86	1.84
	3	8.48	8.52	0.05	25.23	25.57	0.34	41.74	42.62	0.88
	5	10.25	10.28	0.03	30.35	30.83	0.48	41.94	42.76	0.82
1.5	1	6.66	6.72	0.05	19.57	20.14	0.57	32.17	33.57	1.40
	3	7.45	7.49	0.05	22.12	22.48	0.36	36.67	37.47	0.80
	5	7.47	7.51	0.04	22.20	22.53	0.33	36.81	37.55	0.74
	Max	10.25	10.28	0.05	30.35	30.83	0.82	49.05	50.10	2.09
	Avg	8.49	8.53	0.04	25.09	25.59	0.50	40.49	41.69	1.20
	Min	6.66	6.72	0.02	19.57	20.14	0.33	32.17	33.57	0.74

Table EC.6 Average Performance of the RR-BPC in the Robust Setting for BIS Instances with 50 Customers.

Γ	0			1			3			5		
τ/γ	opt (%)	gap(LP%)	cpu (s)	opt (%)	gap(LP%)	cpu (s)	opt (%)	gap(LP%)	cpu (s)	opt (%)	gap(LP%)	cpu (s)
0.5	93	0.50	800.66	87	0.76	1119.69	80	0.88	1681.51	82	0.85	1446.24
1	100	0.35	76.97	100	0.54	265.24	91	0.64	1178.80	91	0.61	1099.10
1.5	100	0.49	86.31	93	0.74	883.39	93	0.85	1043.07	100	0.85	681.65
Avg	98	0.45	321.31	93	0.68	756.11	88	0.79	1301.13	91	0.77	1075.67

In the aftermath of a disaster, debris can degrade road conditions, increasing travel time (Lorca et al. 2017, Avishan et al. 2023). To model this uncertainty, we thereby assume that the vehicle travel times are independent random variables that fall within the range of estimated values, denoted by

$$\tilde{d}_{i,j} = (1+\varepsilon\zeta_{i,j})d_{i,j},\ 0 \leq \zeta_{i,j} \leq 1,\ \forall (i,j) \in \mathcal{A}^1, \tag{EC.18}$$

where $\zeta_{i,j}$ is the scale deviation, and ε is the scalar denoting the *uncertainty level*. Lu (2013) suggest the set $\{0.5, 1.5, 2.5\}$ for the values of ε to model minor, moderate, and severe disruptions, respectively. Notably, setting $\varepsilon = 0$ is equivalent to the deterministic model.

To balance the conservatism and tractability of the robust model, we represent the data uncertainty by the polyhedral uncertainty set, which has been applied to other routing problems with uncertainty (Munari et al. 2019), and is written as

$$\mathcal{U} = \left\{ \tilde{d} \in \mathbb{R}_+^{|\mathcal{A}^1|} \mid \text{(EC.18)}, \sum_{(i,j)\in\mathcal{A}^1} \zeta_{i,j} \leq \Gamma \right\}, \tag{EC.19}$$

where the level of conservatism is bounded by the *budget of uncertainty* Γ (Bertsimas and Sim 2004). In other words, it limits the number of arcs whose travel times are allowed to deviate from their nominal values. In practice, the budget Γ is determined by the decision-maker based on personal judgment. In our experiments, Γ takes values from the set $\{0, 1, 3, 5\}$. Note that we often have about seven arcs traveled by vehicles. Consequently, when $\Gamma = 5$ and $\varepsilon = 2.5$, it represents a scenario where the travel times significantly deviate from the nominal ones. The corresponding robust solution is used to hedge against a severely adverse situation. Furthermore, we show how to adapt the DP algorithm for the robust model in EC.11.1.

For each combination of $\varepsilon = 0.5, 1.5, 2.5$, and $\Gamma = 1, 3, 5$, we run experiments based on BIS Instace-Set-50, which has 45 instances with 50 customers, using the proposed RR-BPC algorithm with the same time limit (7,200 seconds) as before. Note that we use the best feasible solution if an instance does not achieve optimality within the time limit. In our analyses, we report three metrics: (i) the price of robustness (PoR), (ii) the fixed-to-robust relative gap (FRG), and (iii) the fixed robust loss (FRL), whose definitions are given in Section 5.3.

Table EC.5 summarizes the results, with the instances grouped by the cable-trench ratios $\tau/\gamma = 0.5, 1.0,$ and 1.5, dropping subscripts a, c and t to improve readability because the instances have $\gamma_a = \gamma_c = \gamma_t$ and $\tau_a = \tau_c = \tau_t$. The PoR values increase markedly with larger uncertainty budgets. For instance, the maximum PoR rises from 10.25% at $\varepsilon = 0.5$ to 49.05% at $\varepsilon = 2.5$. So, higher costs are associated with more conservative solutions, providing stronger protection against uncertainty. By comparing the PoR and FRG, we observe that the nominal solution is almost always suboptimal in the robust setting. This is evidenced by the average FRG values consistently exceeding the PoR values across all parameter combinations. We further assess the degree to which the quality of these fixed nominal solutions deteriorates under uncertainty. The FRL values indicate that at smaller uncertainty levels ($\varepsilon = 0.5$), the impact of uncertainty on the nominal solution is very minimal, with average values between 0.02% and 0.05%. However, as uncertainty increases, so do the FRL values, reaching a maximum of 2.09% and averaging 1.20% at $\varepsilon = 2.5$. These important findings suggest that although deterministic solutions may be suboptimal in the robust setting, the average loss in quality is marginal. Given this observation, it is reasonable to focus on the deterministic (nominal) case as the first step in solving the GDP and switch to the robust model according to the decision-maker's preferences. Ultimately, the decision should be based on the trade-off between the increased computational effort required for solving robust models and the potential cost savings they deliver when uncertainty is substantial, as in emergency scenarios.

Table EC.6 summarizes the performance of the RR-BPC approach in the robust setting. It presents the percentage of instances solved to optimality under the respective parameters, as well as the average root node LP relaxation gaps and running times, computed over both optimal and non-optimal instances. The results show that increasing the budget Γ generally makes the problem more challenging to solve. Average computational times rise from 321.31 seconds in the nominal case ($\Gamma = 0$) to 1301.13 seconds and 1075.67 seconds for $\Gamma = 3$ and $\Gamma = 5$, respectively. The LP relaxation gaps also increase from 0.45% to 0.68%, 0.79%, and 0.77% for the respective budgets. This may explain why the algorithm takes longer to converge toward optimality. While the RR-BPC optimally solves 98% of instances under the nominal setting, this proportion decreases in robust cases. The worst performance is observed for $\Gamma = 3$ and $\tau/\gamma = 0.5$, where only 80% of instances are solved to optimality within the time limit. Overall, although robust settings increase computational effort, the RR-BPC approach maintains strong performance, effectively finding optimal or near-optimal solutions.

EC.11.1 Modifying the Pricing Problem for Robustness Analysis

Here, we show how the uncertainties are addressed in the DP algorithm presented in Section 4.3.2.

Let ι denote the number of arcs whose travel times have reached their worst-case value in a route, such that $\iota \in \{0, 1, \ldots, \Gamma\}$. To account for these values, the DP states must be redefined as $(i, U, w_0, w_1, \ldots, w_\Gamma)$, where w_ι indicates the arrival time at vertex $i \in \mathcal{V}^-$ when ι travel times deviated from their nominal values. Given state $(i, U, w_0, w_1, \ldots, w_\Gamma)$, let w'_ι be the arrival time at vertex $j \in \mathcal{N}_l$, $\mathcal{N}_l \in \mathcal{N}$ when extending over the arc $(i, j) \in \mathcal{A}^1$. Following the recursive equations proposed in Munari et al. (2019), the value w'_ι is computed for each arc (i, j) as follows:

$$w'_\iota = \begin{cases} w_0 + d_{i,j} + f_j, & \iota = 0, \\ \max\{w_\iota + d_{i,j} + f_j, w_{\iota-1} + (1+\varepsilon)d_{i,j} + f_j\}, & \iota = 1, 2, \ldots, \Gamma. \end{cases} \tag{EC.20}$$

In addition, please note that, since the values w are non-decreasing and the routes are always feasible for any realization of the uncertain parameters, the worst case is always achieved by taking the largest travel time deviations of the arcs visited in the route. Consequently, expression (4) (the smallest reduced cost of visiting customer j immediately after customer i) is updated as

$$\tilde{c}_{i,j}(w) = \min_{q \in Q_j} \left\{ (C_l w'_\Gamma - \eta_j) q \right\},$$

where the value of w'_Γ is computed from the recursive equations (EC.20). The dominance relations must be carefully modified to prevent the algorithm from discarding states that lead to the optimal solution. We say that state $(i, U', w'_0, w'_1, \ldots, w'_\Gamma)$ is dominated by state $(i, U, w_0, w_1, \ldots, w_\Gamma)$ if $\Pi(i, U, w_0, w_1, \ldots, w_\Gamma) \leq \Pi(i, U', w'_0, w'_1, \ldots, w'_\Gamma)$, $U \subseteq U'$ and $w_\iota \leq w'_\iota$, $\forall \iota \in \{0, 1, \ldots, \Gamma\}$.

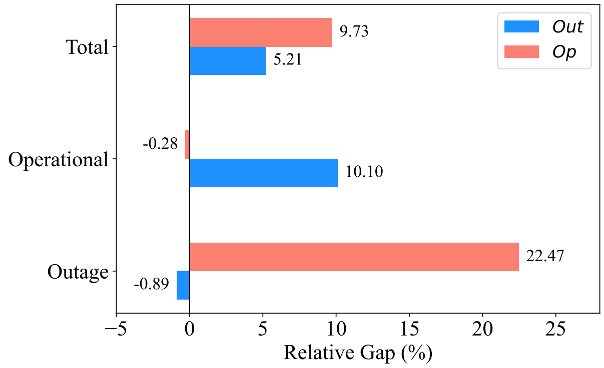
EC.12 Scenario Analyses

This section conducts several analyses. First, we demonstrate the balance between the outage and operational costs in EC.12.1. Next, because the GDP with the RR-BPC approach is instrumental for decision-making, we present an analysis of the impact of fleet size in EC.12.2, and generator number in EC.12.3. Furthermore, we present an analysis of the effect of cluster priority in EC.12.4.

EC.12.1 Balance Between Outage and Operational Costs.

Outage costs reflect the urgency of restoring power to customers as quickly as possible, accounting for penalties associated with customer downtime. Meanwhile, operational costs encompass resource allocation (cable-trench material costs) and logistical challenges (vehicle routing costs). We consider three scenarios: *Op*, *Out*, and *Int*. Scenario *Op* minimizes operational costs while ignoring outage costs; *Out* minimizes outage costs only; and *Int* simultaneously minimizes both costs, serving as the base case. The results are compared across 25 instances selected from Instance-Set-50. For each instance, the RR-BPC algorithm is

Figure EC.3 Value of Integrated Cost Optimization.



Note. Average relative gaps for the optimization scenarios Op and Out are compared with the baseline Int.

used to find optimal solutions under the three scenarios. The total costs for Op and Out are computed by reinserting the outage and operational costs, respectively, into the obtained solutions.

The total costs obtained from scenarios Op and Out are averaged over the 25 selected instances and compared against those of scenario Int. Figure EC.3 summarizes the key findings. Scenario Out reduces outage costs by 0.89% on average, compared to the base case Int, but increases operational costs by 10.10%. Similarly, scenario Op decreases operational costs by 0.28% while increasing outage costs by an average of 22.47%. In contrast, the integrated optimization scenario Int, which minimizes the combined objective function of the two cost components, achieves the lowest total cost. On average, the total costs in Out and Op are 5.21% and 9.73% higher than Int, respectively. More importantly, Int achieves a balance between the two cost components. These findings demonstrate that such integrated optimization offers a more holistic approach for managing outage and operational costs, mitigating the limitations of single-objective approaches.

Proposing a plan that effectively minimizes human suffering and efficiently utilizes available resources is challenging, given the limited funding allocated to disaster response (Besiou and Van Wassenhove 2019, Ye et al. 2020). The above results provide actionable insights for decision-makers. Minimizing operational costs alone (scenario Op) may be helpful in scenarios where material availability is strictly constrained–a situation often encountered in disaster response (Eftekhar et al. 2022). On the other hand, scenario Out, which prioritizes outage costs, is more suitable when minimizing the power restoration time to clustered communities is of utmost importance and the resources, such as cable-trench materials, are in ample supply. However, focusing solely on one perspective can lead to significantly higher total costs.

EC.12.2 Impact of Fleet Size

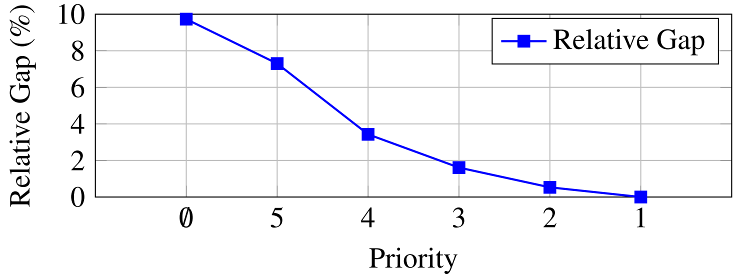
The fleet size is set at the minimum to satisfy all the demand so far. However, it would be helpful to understand the consequences of having extra trucks. To this end, we analyze the impact of fleet expansion on the objective function value, using the same instance set as in Section EC.12.1. The average fleet size in these selected instances is four vehicles. In our computational analyses, we examine the effect of fleet expansion by adding one vehicle at a time until the fleet size doubles, with a maximum of 8 vehicles.

Adding extra vehicles to a fleet can significantly affect the objective function value. The improvements are 4.65%, 7.53%, 9.38%, and 10.61% for one, two, three, and four extra vehicles, respectively. Adding one or two vehicles to the fleet results in notable improvements, with cost reductions of 4.65% and 7.53%, respectively. However, the rate of improvement decreases as more vehicles are added. For example, adding the fourth vehicle results in a 10.61% improvement, which is only a 1.23% increase over the previous improvement of 9.38%, indicating diminishing returns. Initially, adding vehicles provides substantial benefits, but the gains decline progressively with each additional vehicle. Thus, decision-makers must carefully weigh the cost of deploying extra vehicles to an outage area against the incremental improvements they deliver.

EC.12.3 Impact of Generator Number

Here, we conduct additional experiments to examine the impact of increasing the number of available generators. Using the same set of instances as in the previous section, we evaluate the effect of deploying additional

Figure EC.4 Impact of Minimizing Outage Costs Only for Clusters with High Priority.



generators on the objective function value. In the computational study, generators are added incrementally, one at a time, with up to four additional generators considered.

The availability of extra generators enables one or more clusters to be served simultaneously by a larger number of generators. The average improvements in the objective function value are 8.49%, 13.79%, 18.01%, and 21.77% when one, two, three, and four additional generators are deployed, respectively. Similar to the fleet expansion experiments, the rate of improvement decreases as more generators are added. Specifically, the second additional generator yields a further 5.29% improvement relative to the single-generator case, whereas the third and fourth additional generators provide incremental improvements of 4.23% and 3.75%, respectively. A closer examination reveals that these gains are driven primarily by reductions in operational costs, which improve by 13.88%, 22.31%, 29.30%, and 35.41%, respectively. In contrast, the corresponding reductions in outage costs are much smaller, amounting to only 0.31%, 0.51%, 0.48%, and 0.52%. Consequently, although increasing generator availability can substantially enhance overall restoration performance, the benefits are diminishing and arise mainly from improved operational efficiency rather than meaningful reductions in outage costs. From a managerial perspective, these findings suggest that while additional generators can help decision-makers lower operational costs, their direct impact on the service experienced by affected customers is relatively limited.

EC.12.4 Effects of Cluster Prioritization

This section examines the impact of cluster prioritization on the overall objective value. Specifically, given cluster priorities ω_l for $\mathcal{N}_l \in \mathcal{N}$, where $1 \leq \omega_l \leq 5$, we evaluate six settings in which outage costs are minimized for selected priority levels. Let $\mathcal{W}$ denote the set of selected priorities for a given setting. The outage costs for clusters with $\omega_l \in \mathcal{W}$ are included in the objective function, while the outage costs for all other clusters are excluded.

In the first setting, $\mathcal{W} = \emptyset$, we analyze scenario *Op*, which minimizes only operational costs, completely ignoring outage costs. In the second setting, $\mathcal{W} = \{5\}$, we consider only the highest-priority clusters. Then we continue this process by adding each of the remaining priority values, i.e., $\omega_l = 4, 3, 2$, and 1, one at a time. Finally, in the last setting, $\mathcal{W} = \{1, \ldots, 5\}$, all outage costs are accounted for in the objective function, which corresponds to the integrated problem *Int* (base case).

Figure EC.4 summarizes the results, using the same 25 instances as in Section EC.12.1. The objective values for each setting are adjusted by reinserting the outage costs of excluded clusters into the solutions. The average relative gaps are then computed for each setting, compared to the base case (*Int*). The results show that ignoring outage costs entirely ($\mathcal{W} = \emptyset$) increases the average total costs by nearly 10%, indicating the significance of incorporating outage costs into the objective function. Focusing solely on clusters with the highest priority ($\mathcal{W} = \{5\}$) also leads to substantial overall costs, with a relative gap of approximately 7.30%. As more priority indices become included into the set $\mathcal{W}$, the relative gap decreases. Specifically, optimizing outage costs for clusters with priority levels $\omega_l \geq 3$ achieves a reasonable trade-off, with an average relative gap of only 1.61%.

Restoring power to high-priority loads before addressing lower-priority areas is a widely recognized practice, both in industry (National Grid 2024) and academia (Yao et al. 2019, Vries and Wassenhove 2020). During unplanned power outages, decision-makers prioritize restoring power to critical areas, such as medical centers and public safety facilities. Based on these findings, we recommend prioritizing clusters with priority indices in the top 3/5 of the range (i.e., $\omega_l = 3, 4, 5$). This approach effectively balances outage and operational costs. It is crucial to note that for the sake of fairness, decision-makers should adopt such

Table EC.7 Average Gaps (%) and Running Times (s) of the RR-BPC Approach over BIS Instances Solved Optimally.

			Root				Cutting Planes						
Instance	τ/γ	opt	gap(LP)	cpu(LP)	gap(LPC)	cpu(LPC)	GCI	SRI(T)	SRI(R)	CLI	MWI	node	cpu(IP)
Set-50	0.5	14	0.35	0.88	0.12	273.37	1.0	3.1	9.0	15.9	68.5	53.6	343.56
	1.0	15	0.35	0.81	0.10	11.01	1.4	4.7	10.0	25.0	37.2	39.2	76.97
	1.5	15	0.49	0.71	0.14	8.90	1.9	5.0	12.0	23.5	21.9	91.1	86.31
#/Avg		44	0.40	0.80	0.12	93.77	1.5	4.3	10.4	21.6	42.0	61.5	164.98
Set-75	0.5	12	0.73	4.15	0.07	10.28	4.2	17.3	9.9	0.8	6.1	40.1	409.43
	1.0	11	1.04	3.53	0.06	8.11	6.5	19.2	14.3	20.6	11.3	26.4	194.90
	1.5	9	1.32	3.60	0.08	9.40	8.8	16.1	11.7	24.0	9.6	42.2	259.42
#/Avg		32	1.00	3.76	0.07	9.26	6.3	17.6	11.9	14.1	8.8	36.0	293.49
Set-100	0.5	6	1.53	8.02	0.08	24.90	17.5	55.2	27.8	6.7	12.0	82.5	2287.43
	1.0	6	1.82	10.77	0.04	22.31	14.5	28.2	19.7	2.2	28.2	98.3	1748.85
	1.5	8	2.43	8.97	0.05	20.64	17.5	41.5	17.8	36.6	28.0	110.1	2339.39
#/Avg		20	1.98	9.25	0.06	22.62	16.6	41.6	21.4	17.3	23.3	98.3	2146.64

Table EC.8 Average Gaps (%) and Running Times (s) of the RR-BPC Approach over Non-Optimal Instances.

			Root				Cutting Planes						
Instance	τ/γ	non-opt	gap(LP)	CPU(LP)	gap(LPC)	CPU(LPC)	GCI	SRI(T)	SRI(R)	CLI	MWI	node	gap(TL)
Set-50	0.5	1	2.66	1.88	1.17	4.66	7.0	38.0	12.0	315.0	696.0	2807.0	0.50
Set-75	0.5	3	1.93	4.15	0.33	10.28	12.0	8.7	31.3	82.7	76.0	965.3	0.13
	1.0	4	1.61	3.53	0.28	8.11	7.5	10.5	37.0	60.3	104.3	1356.0	0.06
	1.5	6	1.80	3.60	0.38	9.40	4.5	22.3	29.7	112.3	58.2	1503.5	0.21
Set-100	0.5	9	1.47	8.02	0.31	24.90	11.6	33.7	26.4	37.2	55.7	430.4	0.22
	1.0	9	1.53	10.77	0.40	22.31	13.7	34.9	33.4	74.4	81.7	363.0	0.30
	1.5	7	0.61	8.97	0.12	20.64	5.6	9.3	18.4	58.6	66.9	855.0	0.04
#/Avg		39	1.46	6.62	0.33	19.79	9.4	23.6	28.2	74.2	87.0	853.2	0.19

prioritization strategies that minimize inequity (Eisenhandler and Tzur 2019, Li et al. 2024, Breugem et al. 2025).

EC.13 Summary of Computational Results

This section first summarizes the performance of the RR-BPC approach. Next, it provides the detailed results.

EC.13.1 Performance of the RR-BPC Approach.

To evaluate the performance of the RR-BPC approach, we apply its best configuration, ‘All’, across all instances. We first look at the instances that are solved optimally. The results are summarized in Table EC.7. The table presents the performance analysis across three BIS instance sets: Set-50, Set-75, and Set-100. These instances are further grouped by the cable-trench ratios $\tau/\gamma = 0.5, 1.0$, and 1.5, where subscripts a, c and t are dropped to improve readability because the instances have $\gamma_a = \gamma_c = \gamma_t$ and $\tau_a = \tau_c = \tau_t$.

The number of instances solved to optimality demonstrates the strong performance of the RR-BPC approach, solving 44, 32, and 20 instances in Sets 50, 75, and 100, respectively—amounting to approximately 71% of all instances. However, as the problem size increases, the number of instances solved optimally decreases. This trend is further evidenced by the root node gap values (gap(LP)), which grow from 0.40% (Instance-Set-50) to 1.00% (Instance-Set-75) and to 1.98% (Instance-Set-100). Table EC.7 reports a large number of separated inequalities GCI, SRI, CLI, and MWI, where (T) and (R) refer to the tree and route variables, respectively. Incorporating these inequalities significantly reduces the relaxation gaps (gap(LPC)), bringing them to 0.12%, 0.07%, and 0.06% for the instances solved optimally in Instance-Set-50, -75, and -100, respectively. The average computation times (‘cpu(IP)’) increase progressively from about 165 seconds for Instance-Set-50 to approximately 294 seconds for Instance-Set-75 and to 2146.64 seconds for Instance-Set-100. This increase correlates with the significant growth in the number of branch-and-bound nodes—61.5 nodes in Instance-Set-50, compared to 98.3 nodes in Instance-Set-100. Interestingly, we observe that instances with the smallest cable-trench ratio, $\tau/\gamma = 0.5$, pose the most significant computational challenges in terms of average running time across all instance sets, at 1,013.47 seconds.

Next, we examine instances not solved to optimality within the TL. With a larger customer number, the problem becomes increasingly challenging. The algorithm optimally solves about 44% of instances with 100 customers. As summarized in Table EC.8, a total of 39 instances across Instance-Set-50, -75, and -100 cannot converge to optimality within the 2-hour time limit. Note that we do not include the results in

Instance-Set-50 with $\tau/\gamma = 1.0$ and 1.5 since all of the instances in these groups are solved optimally. The root node gap (gap(LP)) averages 1.46%, reflecting a relatively looser relaxation. However, the cutting planes significantly reduce this gap at the root node, as demonstrated by the gap(LPC) values, which average 0.33%. Despite this tightening, the branch-and-bound exploration (node count in Table EC.8) averages 853.2 nodes, indicating a significant increase in the total number of branch-and-bound nodes compared to the instances solved optimally. It is worth noting that the average gap is 0.19% for the unsolved instances. Overall, it is safe to conclude that the RR-BPC approach efficiently obtains optimal or near-optimal solutions.

EC.13.2 Detailed Computational Results

This section reports detailed computational results on solving basic instances (set BIS) by the solver (Table EC.9) and the proposed exact algorithm RR-BPC (Tables EC.10, EC.11, and EC.12). Furthermore, Table EC.13 reports the computational results of running the RR-BPC algorithm to solve instance set VST used in the analyses in Section 5.4. The above tables group the instances by the cable-trench ratio τ/γ, where $\gamma_a = \gamma_c = \gamma_t$ and $\tau_a = \tau_c = \tau_t$. Lastly, Table EC.14 summarizes the computational results for instance set VTC with varying cable-trench coefficients (i.e., $\gamma_a \neq \gamma_c \neq \gamma_t$ and $\tau_a \neq \tau_c \neq \tau_t$).

The column ‘Instance’ provides the instance names. The columns ‘H_0’ and ‘F_R’ report the relative gaps and computational times for the upper-bounding algorithm and the algorithm for finding feasible solutions based on a partial solution, respectively. The column ‘call’ indicates the number of times the corresponding algorithm is invoked. If F_R is not invoked by the main algorithm or all the returned solutions are infeasible, the table reports ‘n/a’. For the non-optimal instances, the columns ‘gap’ report the gaps between the best feasible integral solution found by the corresponding algorithm and the best-known solution. Additionally, the column ‘CPU(IP)’ reports TL$(\cdot)$ for non-optimal instances, indicating the relative gap at termination. The optimal (resp., best-known) objective values are reported under the column ‘Z_{F_2}’ for the optimal (resp., non-optimal) instances. Lastly, the column ‘CFI’ reports the average number of the capacity feasibility inequalities (Section 4.4). The remaining columns are defined as in the previous experiments.

Table EC.9 Detailed Results on Solving Formulation F_1 by CPLEX.

Instance	Solver				RR-BPC	
	Z_{F_1}	gap(LP)	gap(TL)	gap(MIP)	CPU	Z_{F_2}
$\tau/\gamma = 0.5$						
ascension50	73,670.59	31.43	0.96	22.86	2.06	72,967.96
beaufort50	71,849.72	43.13	2.29	32.09	7.12	70,237.89
brazoria50	86,941.55	40.84	7.97	38.41	372.85	80,527.08
brevard50	67,375.99	51.07	1.09	40.97	355.32	66,650.34
brunswick50	83,877.63	45.53	0.13	35.06	178.13	83,772.33
harris50	107,526.31	37.63	1.91	30.89	0.65	105,508.02
jefferson50	95,561.67	35.81	1.03	26.14	60.02	94,586.10
lafourche50	96,466.37	42.64	2.30	29.44	TL(0.50)	94,299.02
lee50	89,406.09	40.68	0.00	31.78	20.94	89,406.09
manatee50	103,812.70	31.37	0.29	23.02	0.64	103,507.49
montgomery50	110,542.03	43.60	1.88	37.04	5.78	108,507.00
palmbeach50	72,871.80*	37.04	28.04	n/a	3800.36	101,264.93
robeson50	102,426.36	55.47	2.41	46.45	1.27	100,015.49
sampson50	80,292.78	46.40	0.32	37.88	4.11	80,039.98
wayne50	87,979.52	44.55	0.00	32.01	0.62	87,979.52
$\tau/\gamma = 1.0$						
ascension50	88,942.05	26.76	2.43	20.63	110.21	86,831.97
beaufort50	80,257.73	38.31	0.13	26.36	7.33	80,151.89
brazoria50	95,991.60	35.66	1.76	30.27	509.68	94,330.31
brevard50	77,183.76	45.87	2.84	38.18	170.01	75,052.66
brunswick50	97,944.90	40.21	1.23	31.43	188.56	96,753.55
harris50	128,949.37	32.19	3.45	27.66	0.59	124,645.73
jefferson50	111,806.89	30.83	0.59	22.02	29.00	111,150.22
lafourche50	106,656.33	35.71	1.56	21.24	24.17	105,014.99
lee50	99,052.87	31.28	0.24	22.89	2.86	98,814.93
manatee50	126,494.80	27.07	1.73	20.65	88.79	124,341.19
montgomery50	121,147.45	35.55	1.93	29.4	1.88	118,855.01
palmbeach50	113,950.23	27.41	0.48	19.96	0.52	113,400.34
robeson50	112,235.44	50.25	1.04	40.95	1.29	111,079.59
sampson50	96,161.03	41.16	5.14	36.47	18.86	91,458.24
wayne50	99,762.25	39.66	0.00	28.28	0.78	99,762.25
$\tau/\gamma = 1.5$						
ascension50	100,728.93	23.47	0.00	16.02	555.60	100,728.93
beaufort50	90,202.55	34.59	0.23	23.98	9.68	89,998.68
brazoria50	106,601.16	28.80	2.91	24.66	13.65	103,586.84
brevard50	83,563.33	41.70	0.35	32.45	49.02	83,269.56
brunswick50	113,150.64	36.25	3.12	29.32	153.46	109,728.38
harris50	150,745.64	28.65	4.20	25.15	11.18	144,674.07
jefferson50	123,395.37	24.66	0.00	15.75	63.16	123,395.37
lafourche50	121,086.46	32.35	0.03	17.01	28.55	121,051.58
lee50	115,527.39	27.51	1.21	20.85	2.43	114,146.32
manatee50	144,390.33	21.75	2.41	16.61	0.72	140,991.36
montgomery50	141,340.31	31.75	4.86	28.24	1.91	134,792.05
palmbeach50	133,943.00	23.94	1.28	17.98	0.59	132,244.00
robeson50	124,765.56	46.73	0.72	37.87	345.74	123,878.30
sampson50	100,987.98	34.09	2.58	28.13	58.13	98,444.82
wayne50	111,819.45	35.93	0.07	25.85	0.80	111,742.05

* The value is the best lower bound obtained at termination due to the absence of a feasible integer solution.

Table EC.10 Detailed Results on BIS Instance-Set-50.

	Root				H_0		F_R			Total								
Instance	gap(LP)	CPU(LP)	gap(LPC)	CPU(LPC)	gap	CPU	gap	call	CPU	GCI	SRI(T)	SRI(R)	CLI	MWI	CFI	node	CPU(IP)	Z_{F_2}
$\tau/\gamma = 0.5$																		
ascension50	0.00	1.09	0.00	1.18	0.56	0.43	n/a	0	0.00	0	0	0	0	0	0	1	2.06	72,967.96
beaufort50	0.17	1.22	0.05	1.43	0.12	0.13	n/a	0	0.00	0	0	15	0	0	0	6	7.12	70,237.89
brazoria50	0.23	1.27	0.12	10.85	0.24	2.16	0.00	21	35.68	0	0	64	105	139	0	322	372.85	80,527.08
brevard50	1.06	1.23	0.15	3.55	0.72	0.12	0.40	7	0.44	14	36	0	0	5	0	40	355.32	66,650.34
brunswick50	0.90	1.22	0.87	1.46	0.08	0.09	0.06	19	0.95	0	2	32	0	4	0	236	178.13	83,772.33
harris50	0.00	0.39	0.00	0.42	0.00	0.06	n/a	0	0.00	0	0	0	0	0	0	1	0.65	105,508.02
jefferson50	0.46	0.72	0.38	1.15	0.15	0.03	0.00	12	0.20	0	0	15	0	6	0	119	60.02	94,586.10
lafourche50	2.66	1.88	1.17	4.66	0.06	0.19	0.00	139	29.80	7	38	12	315	696	1213	2807	TL(0.50)	94,299.02
lee50	0.38	0.77	0.12	1.02	0.00	1.59	0.00	4	6.25	0	6	0	1	9	0	19	20.94	89,406.09
manatee50	0.00	0.30	0.00	0.32	0.09	0.13	n/a	0	0.00	0	0	0	0	0	0	1	0.64	103,507.49
montgomery50	0.13	1.16	0.00	3.50	0.66	1.83	n/a	0	0.00	0	0	0	8	10	0	1	5.78	108,507.00
palmbeach50	1.56	0.51	0.00	3799.42	0.00	0.54	n/a	0	0.00	0	0	0	108	786	0	1	3800.36	101,264.93
robeson50	0.00	0.91	0.00	0.91	0.10	0.20	n/a	0	0.00	0	0	0	0	0	0	1	1.27	100,015.49
sampson50	0.00	1.22	0.00	1.67	0.62	1.79	n/a	0	0.00	0	0	0	0	0	0	1	4.11	80,039.98
wayne50	0.00	0.34	0.00	0.34	0.00	0.03	n/a	0	0.00	0	0	0	0	0	0	1	0.62	87,979.52
$\tau/\gamma = 1.0$																		
ascension50	0.04	1.02	0.04	1.10	0.55	0.42	0.00	6	1.15	0	0	9	2	4	42	163	110.21	86,831.97
beaufort50	0.15	1.09	0.03	1.96	0.11	0.09	n/a	0	0.00	0	0	11	0	1	0	6	7.33	80,151.89
brazoria50	0.67	0.69	0.05	141.09	0.21	60.00	n/a	6	60.00	0	0	40	123	334	0	29	509.68	94,330.31
brevard50	1.07	2.03	0.15	3.92	0.58	0.19	0.34	5	0.54	14	16	8	0	9	0	48	170.01	75,052.66
brunswick50	0.79	1.49	0.67	1.95	0.18	0.07	0.03	18	0.70	2	28	39	0	6	0	200	188.56	96,753.55
harris50	0.00	0.31	0.00	0.34	0.00	0.08	n/a	0	0.00	0	0	0	0	0	0	1	0.59	124,645.73
jefferson50	0.39	0.65	0.31	1.04	0.13	0.03	0.00	4	0.06	0	7	11	0	29	0	39	29.00	111,150.22
lafourche50	1.57	1.11	0.11	1.63	0.02	0.03	0.00	3	0.04	5	17	0	0	0	0	21	24.17	105,014.99
lee50	0.09	0.54	0.09	0.55	0.00	0.04	0.00	1	0.02	0	3	0	0	0	0	7	2.86	98,814.93
manatee50	0.42	0.37	0.10	3.55	0.07	0.46	0.00	11	4.24	0	0	32	137	62	0	69	88.79	124,341.19
montgomery50	0.03	0.38	0.00	1.32	0.62	0.15	n/a	0	0.00	0	0	0	1	1	0	1	1.88	118,855.01
palmbeach50	0.00	0.31	0.00	0.31	0.00	0.00	n/a	0	0.00	0	0	0	0	0	0	1	0.52	113,400.34
robeson50	0.00	0.77	0.00	0.78	0.05	0.33	n/a	0	0.00	0	0	0	0	0	0	1	1.29	111,079.59
sampson50	0.05	0.86	0.00	5.16	0.55	13.20	n/a	0	0.00	0	0	0	112	112	0	1	18.86	91,458.24
wayne50	0.00	0.46	0.00	0.46	0.00	0.03	n/a	0	0.00	0	0	0	0	0	0	1	0.78	99,762.25
$\tau/\gamma = 1.5$																		
ascension50	0.09	0.80	0.06	1.72	0.46	0.25	0.00	12	1.88	0	0	10	12	22	263	801	555.60	100,728.93
beaufort50	0.13	1.50	0.03	2.37	0.09	0.09	n/a	0	0.00	0	0	11	0	1	0	6	9.68	89,998.68
brazoria50	0.43	0.55	0.07	4.92	0.19	0.26	0.00	1	0.31	0	0	22	101	20	0	7	13.65	103,586.84
brevard50	0.95	1.03	0.15	2.33	0.39	0.88	0.29	2	1.24	14	0	8	0	26	0	35	49.02	83,269.56
brunswick50	0.71	1.38	0.60	1.94	0.06	0.06	0.02	12	0.39	2	28	42	0	8	0	164	153.46	109,728.38
harris50	0.61	0.25	0.10	1.09	0.00	0.19	n/a	2	0.29	0	0	23	118	32	0	15	11.18	144,674.07
jefferson50	0.47	0.47	0.41	0.91	0.10	0.01	0.00	12	0.08	0	0	16	2	19	0	209	63.16	123,395.37
lafourche50	1.45	0.99	0.09	1.28	0.03	0.03	0.00	3	0.04	13	47	0	0	5	0	17	28.55	121,051.58
lee50	0.07	0.38	0.07	0.38	0.00	0.04	0.00	1	0.02	0	0	0	0	0	0	7	2.43	114,146.32
manatee50	0.04	0.22	0.00	0.43	0.09	0.08	n/a	0	0.00	0	0	0	0	4	0	1	0.72	140,991.36
montgomery50	0.02	0.38	0.00	1.30	0.56	0.12	n/a	0	0.00	0	0	0	2	2	0	1	1.91	134,792.05
palmbeach50	0.16	0.27	0.00	0.37	0.03	0.03	n/a	0	0.00	0	0	0	2	2	0	1	0.59	132,244.00
robeson50	1.40	1.56	0.03	111.05	0.04	1.63	n/a	0	0.00	0	0	14	114	164	0	6	345.74	123,878.30
sampson50	0.57	0.45	0.48	2.76	0.41	0.62	n/a	4	1.24	0	0	34	0	22	0	96	58.13	98,444.82
wayne50	0.18	0.41	0.00	0.58	0.00	0.02	n/a	0	0.00	0	0	0	2	2	0	1	0.80	111,742.05

Table EC.11 Detailed Results on BIS Instance-Set-75.

	Root				H_0		F_R			Total								
Instance	gap(LP)	CPU(LP)	gap(LPC)	CPU(LPC)	gap	CPU	gap	call	CPU	GCI	SRI(T)	SRI(R)	CLI	MWI	CFI	node	CPU(IP)	Z_{F_2}
$\tau/\gamma = 0.5$																		
craven75	0.62	5.79	0.00	6.48	0.13	0.48	n/a	0	0.00	8	0	0	0	0	0	1	7.88	108,028.26
dorchester75	0.41	4.87	0.19	6.36	0.02	0.14	0.00	3	0.22	0	8	3	0	11	0	33	133.52	137,188.99
duplin75	1.24	5.37	0.25	5.68	0.05	0.24	0.00	34	4.57	6	0	0	139	155	589	1507	TL(0.05)	115,790.49
eastbatonrouge75	1.81	3.65	0.40	15.01	0.37	0.74	0.06	169	60.00	27	0	51	0	8	0	920	TL(0.10)	107,596.71
galveston75	0.46	5.15	0.06	9.07	0.26	1.18	0.04	2	1.98	4	16	0	1	1	0	5	44.56	114,951.99
hidalgo75	0.80	5.93	0.00	9.62	0.01	0.32	n/a	0	0.00	2	5	0	2	4	0	1	11.81	149,670.20
lafayette75	2.73	6.92	0.34	28.79	0.10	0.22	0.00	20	4.42	3	26	43	109	65	2	469	TL(0.23)	116,886.91
livingston75	0.14	2.41	0.10	3.57	0.19	0.72	0.00	2	0.98	1	2	0	0	0	0	9	22.78	119,313.63
orleans75	0.82	4.69	0.23	9.95	0.07	0.16	0.00	7	1.01	10	0	39	1	27	38	268	1350.60	112,982.57
pender75	3.45	4.15	0.12	41.16	0.00	0.98	n/a	2	2.76	19	80	21	0	15	0	25	970.35	143,966.13
pinellas75	1.51	1.96	0.06	7.64	0.19	0.38	0.00	8	3.11	4	76	3	0	0	22	119	2229.46	112,841.85
pitt75	0.02	1.55	0.00	3.26	0.22	0.38	n/a	0	0.00	0	0	0	0	3	0	1	6.20	131,251.34
sttammany75	0.22	6.39	0.05	13.60	0.04	0.87	0.00	2	1.87	1	13	12	0	5	4	13	68.96	128,001.41
terrebonne75	0.18	6.21	0.01	11.67	0.08	0.96	n/a	0	0.00	1	8	38	0	2	0	3	59.86	111,068.54
volusia75	0.13	0.69	0.05	0.98	0.03	0.12	0.00	1	0.05	0	0	3	5	5	0	3	7.14	174,987.58
$\tau/\gamma = 1.0$																		
craven75	0.96	4.91	0.00	5.71	0.11	0.45	n/a	0	0.00	16	0	0	0	0	0	1	6.93	126,314.91
dorchester75	0.25	9.91	0.16	10.58	0.02	0.15	0.00	3	0.29	0	8	3	0	44	0	23	167.23	161,129.15
duplin75	1.21	4.52	0.23	9.48	0.11	0.08	0.00	87	4.33	6	0	38	133	369	558	1278	TL(0.06)	135,549.47
eastbatonrouge75	1.49	4.69	0.36	20.55	0.35	0.70	0.07	133	55.17	18	0	47	0	13	0	923	TL(0.02)	113,941.38
galveston75	0.97	8.46	0.08	11.84	0.21	4.44	0.01	2	4.99	4	15	0	1	1	0	7	127.47	123,709.70
hidalgo75	1.07	2.44	0.03	5.64	0.00	0.24	n/a	0	0.00	13	17	8	0	8	0	11	66.78	165,322.13
lafayette75	3.26	3.35	0.22	7.04	0.13	0.05	0.00	23	0.59	5	29	34	3	8	4	1705	TL(0.08)	132,630.56
livingston75	0.55	2.17	0.11	9.63	0.00	1.83	n/a	5	4.68	1	8	79	114	21	0	77	393.73	138,123.39
orleans75	0.80	2.42	0.06	3.24	0.11	0.11	0.00	1	0.05	11	0	2	0	0	0	13	50.72	129,626.30
pender75	4.10	2.02	0.03	22.80	0.00	0.22	n/a	1	0.10	20	73	10	0	3	0	3	118.90	155,853.90
pinellas75	1.71	1.94	0.05	9.37	0.15	0.33	0.00	7	1.64	4	74	3	0	5	16	61	1029.14	133,590.03
pitt75	0.08	0.97	0.02	4.56	0.17	1.82	0.01	1	0.42	0	0	6	0	12	2	24	38.43	154,600.18
sttammany75	0.48	4.77	0.31	10.65	0.07	0.18	0.00	41	3.01	1	13	29	105	27	450	1518	TL(0.08)	149,530.37
terrebonne75	0.63	2.57	0.03	3.23	0.07	0.16	0.00	1	0.08	3	16	27	0	0	2	17	59.34	124,990.44
volusia75	0.29	1.05	0.07	2.58	0.03	0.05	0.00	5	0.13	0	0	19	112	30	0	53	85.26	206,296.53
$\tau/\gamma = 1.5$																		
craven75	1.70	5.87	0.12	18.21	0.07	0.13	0.00	20	2.05	16	16	36	121	14	230	541	TL(0.05)	145,206.26
dorchester75	0.72	4.58	0.47	19.11	0.01	0.07	0.00	82	1.81	0	0	7	112	50	356	1100	TL(0.23)	186,036.27
duplin75	0.77	2.54	0.03	3.69	0.02	0.08	0.00	2	0.08	7	0	0	2	2	0	7	29.69	151,028.53
eastbatonrouge75	1.33	4.11	0.29	14.49	0.34	1.50	0.08	8	5.27	16	0	36	0	15	0	36	585.30	120,225.71
galveston75	1.41	8.59	0.09	17.97	0.19	1.39	0.00	2	2.68	4	15	0	102	3	0	9	202.88	132,437.96
hidalgo75	1.57	3.06	0.02	6.14	0.06	0.14	0.04	2	0.16	13	22	11	4	15	0	12	68.28	186,394.25
lafayette75	4.05	6.02	0.30	12.77	0.08	1.00	0.00	24	3.61	5	29	19	87	10	114	1375	TL(0.17)	153,022.65
livingston75	1.60	2.31	0.71	22.68	0.00	0.11	n/a	96	6.97	2	11	64	130	158	131	1792	TL(0.42)	157,925.46
orleans75	0.87	3.93	0.03	5.89	0.08	0.11	n/a	0	0.00	14	0	2	0	2	0	5	36.20	150,624.92
pender75	4.49	2.41	0.06	23.48	0.03	0.25	0.00	3	0.40	23	96	9	0	11	0	8	308.92	174,151.71
pinellas75	2.36	1.98	0.46	16.76	0.17	0.07	0.00	52	2.36	4	78	33	104	24	24	273	TL(0.38)	154,963.22
pitt75	0.37	1.08	0.24	8.23	0.14	0.06	0.00	43	1.45	0	0	19	120	93	779	3940	TL(0.01)	178,370.68
sttammany75	0.20	3.39	0.10	7.13	0.05	0.38	0.00	9	2.30	0	0	24	0	20	28	233	833.18	165,524.29
terrebonne75	1.08	3.59	0.05	4.13	0.05	0.16	0.00	5	0.44	2	12	23	0	1	6	65	262.85	143,044.73
volusia75	0.12	0.76	0.03	1.68	0.02	0.03	0.00	1	0.01	0	0	0	108	17	0	5	7.46	231,458.23

Table EC.12 Detailed Results on BIS Instance-Set-100.

	Root				H_0		F_R			Total								
Instance	gap(LP)	CPU(LP)	gap(LPC)	CPU(LPC)	gap	CPU	gap	call	CPU	GCI	SRI(T)	SRI(R)	CLI	MWI	CFI	node	CPU(IP)	Z_{F_2}
$\tau/\gamma = 0.5$																		
berkeley100	1.18	14.86	0.45	47.86	0.19	1.68	0.00	31	9.81	6	33	0	142	177	48	153	TL(0.40)	192,060.56
broward100	0.73	8.13	0.02	23.71	0.04	1.06	n/a	0	0.00	13	30	3	39	31	0	3	155.17	208,494.84
calcasieu100	0.85	16.62	0.15	40.18	0.03	0.21	0.00	18	2.26	32	73	9	0	12	0	189	3488.80	180,322.73
cameron100	0.42	13.46	0.34	25.85	0.05	2.74	0.00	45	59.04	5	23	51	0	24	0	287	TL(0.14)	189,256.29
collier100	4.54	5.45	0.12	43.41	0.01	0.31	0.00	5	0.90	31	122	41	0	3	0	55	4034.23	201,063.91
fortbend100	0.36	5.31	0.29	9.45	0.18	10.10	0.00	18	54.19	0	0	40	1	75	258	360	TL(0.21)	267,021.94
galveston100	1.45	7.38	0.31	16.40	0.04	0.96	0.00	45	28.00	8	26	8	105	15	240	417	TL(0.22)	221,822.73
jefferson100	1.41	13.95	0.11	23.67	0.13	0.93	0.00	15	9.07	28	26	0	0	17	80	237	TL(0.06)	151,144.38
lafayette100	0.68	25.99	0.07	46.62	0.52	0.33	0.11	27	2.68	15	80	60	0	3	0	172	4914.79	101,252.01
miamidade100	2.21	5.92	0.04	9.13	0.10	0.55	0.00	2	1.08	13	13	4	0	4	0	9	123.68	163,706.16
nueces100	6.19	8.44	0.14	37.69	0.01	0.56	0.00	20	7.32	26	105	20	0	8	112	144	TL(0.11)	174,901.59
pasco100	1.30	11.84	0.70	20.92	0.00	0.32	n/a	13	3.21	4	40	3	48	39	138	327	TL(0.63)	183,792.64
polk100	0.18	4.00	0.09	8.09	0.06	0.93	0.01	8	5.47	1	13	50	1	19	0	67	1007.89	207,395.47
sarasota100	0.49	4.53	0.39	16.01	0.07	2.71	0.00	290	26.34	0	0	50	6	118	12	1267	TL(0.19)	214,280.01
stlucie100	0.48	8.11	0.10	19.78	0.00	0.63	n/a	43	11.65	27	50	66	33	28	38	682	TL(0.01)	191,376.48
$\tau/\gamma = 1.0$																		
berkeley100	0.89	8.23	0.32	39.66	0.22	17.46	0.00	20	20.02	6	22	22	107	136	112	252	TL(0.28)	223,541.75
broward100	0.81	11.28	0.08	19.65	0.04	0.33	0.00	22	2.40	14	20	23	0	17	56	415	6924.40	242,033.70
calcasieu100	1.19	14.32	0.31	48.75	0.02	1.02	0.00	62	43.04	24	40	29	68	40	0	253	TL(0.21)	205,589.68
cameron100	1.22	8.96	0.90	25.17	0.05	6.01	0.00	15	11.32	2	17	81	103	86	30	157	TL(0.82)	220,275.82
collier100	4.90	4.83	0.16	17.91	0.01	0.27	0.00	30	1.16	29	80	34	29	35	32	386	TL(0.03)	224,761.35
fortbend100	0.14	5.45	0.04	12.59	0.15	0.46	0.00	1	0.22	0	0	39	2	109	6	31	428.27	311,266.30
galveston100	1.76	6.00	0.42	27.39	0.07	0.51	0.00	107	43.44	10	40	32	0	59	32	666	TL(0.28)	257,387.35
jefferson100	1.80	13.30	0.20	30.65	0.11	1.09	0.00	55	38.49	25	71	31	28	41	204	504	TL(0.13)	174,641.81
lafayette100	0.66	28.78	0.06	41.94	0.49	0.81	n/a	9	4.45	17	40	44	0	2	0	121	2652.68	111,335.46
miamidade100	2.45	5.71	0.03	11.56	0.20	0.87	0.00	2	1.21	13	0	0	0	29	0	9	137.44	194,750.06
nueces100	6.31	4.17	0.00	33.25	0.08	0.21	n/a	0	0.00	31	102	12	8	9	0	1	36.11	202,698.04
pasco100	0.53	9.21	0.03	14.88	0.01	0.52	0.00	1	0.32	12	7	0	3	3	0	13	314.22	205,168.18
polk100	0.57	4.14	0.47	12.01	0.05	0.13	0.00	48	4.25	1	13	14	118	102	144	340	TL(0.36)	240,059.25
sarasota100	0.76	3.77	0.52	18.92	0.00	0.20	n/a	118	31.05	0	0	24	115	136	0	477	TL(0.38)	248,075.25
stlucie100	0.71	8.21	0.28	21.00	0.08	0.25	0.00	42	6.94	26	31	34	102	100	0	232	TL(0.22)	226,098.20
$\tau/\gamma = 1.5$																		
berkeley100	0.70	5.81	0.09	23.14	0.16	4.26	0.00	23	5.76	6	0	29	0	121	220	617	TL(0.04)	255,748.50
broward100	1.02	7.07	0.12	16.79	0.05	0.17	0.00	46	7.12	10	26	0	1	47	534	826	TL(0.06)	280,133.10
calcasieu100	1.00	7.31	0.09	22.00	0.06	0.32	0.00	15	2.18	24	40	27	1	24	0	146	4741.19	222,854.90
cameron100	0.16	8.57	0.03	16.15	0.04	0.75	0.00	4	1.79	4	25	11	67	68	6	79	1443.19	241,989.48
collier100	5.21	4.71	0.12	13.54	0.01	0.11	0.00	5	0.26	26	80	4	35	28	0	123	3485.09	255,179.33
fortbend100	0.14	4.88	0.04	22.20	0.15	0.25	0.00	7	0.75	1	3	32	178	65	133	454	TL(0.00)	357,239.97
galveston100	1.38	5.15	0.05	17.81	0.08	0.42	0.00	11	2.27	5	40	25	0	41	0	120	1994.63	285,601.83
jefferson100	1.82	6.33	0.03	15.48	0.13	0.40	0.00	6	1.61	19	41	13	114	9	0	48	993.97	192,233.09
lafayette100	0.71	28.58	0.06	40.46	0.46	0.76	n/a	19	9.75	23	40	53	0	2	0	251	5176.12	121,404.30
miamidade100	2.74	5.04	0.06	9.53	0.06	0.32	0.00	9	1.04	13	6	0	72	29	4	113	847.84	222,061.90
nueces100	6.44	6.10	0.00	30.12	0.00	0.16	n/a	0	0.00	26	60	9	4	23	0	1	33.06	234,077.21
pasco100	0.95	10.11	0.10	19.97	0.01	0.22	0.00	8	1.16	11	7	0	113	34	129	321	TL(0.03)	233,516.56
polk100	0.51	3.90	0.17	7.93	0.03	0.20	0.00	37	5.84	2	11	36	0	43	394	1381	TL(0.05)	266,210.67
sarasota100	0.31	4.30	0.20	11.87	0.04	0.16	0.00	142	15.02	0	0	27	116	110	186	1359	TL(0.07)	274,755.51
stlucie100	0.63	5.60	0.14	11.06	0.02	0.11	0.00	110	9.02	9	18	5	2	48	274	1027	TL(0.05)	254,433.72

Table EC.13 Detailed Results on VST Instance-Set-50 with Varying Material Unloading Times Used in Section 5.4.

	Root				H_0		F_R			Total								
Instance	gap(LP)	CPU(LP)	gap(LPC)	CPU(LPC)	gap	CPU	gap	call	CPU	GCI	SRI(T)	SRI(R)	CLI	MWI	CFI	node	CPU(IP)	Z_{F_2}
$\tau/\gamma = 0.5$																		
ascension50-0.5	0.03	1.88	0.00	2.21	6.07	0.15	n/a	0	0.00	0	0	0	0	1	0	1	3.05	82,991.70
beaufort50-0.5	0.37	1.59	0.29	1.95	15.66	0.09	0.00	4	0.10	0	0	8	0	0	0	50	30.99	76,038.67
brazoria50-0.5	0.07	0.65	0.00	2.92	5.16	0.18	n/a	0	0.00	0	0	32	0	100	0	1	4.17	92,102.99
brevard50-0.5	1.76	1.23	0.16	46.26	13.13	0.05	n/a	0	0.00	17	78	11	0	4	0	7	389.93	74,311.53
brunswick50-0.5	1.05	1.26	0.96	1.96	7.37	0.06	0.00	43	1.15	0	0	37	0	5	0	638	562.57	93,160.41
harris50-0.5	0.00	0.57	0.00	0.61	3.04	0.04	n/a	0	0.00	0	0	0	0	0	0	1	0.91	117,046.79
jefferson50-0.5	0.53	0.77	0.42	1.31	7.27	0.04	0.00	3	0.04	4	16	14	0	8	0	29	32.37	104,471.59
lafourche50-0.5	4.16	7.83	1.73	21.38	15.34	0.10	0.00	140	54.90	31	70	22	46	185	158	1430	TL(1.01)	103,364.42
lee50-0.5	1.09	0.54	0.86	4.36	2.47	0.08	0.00	53	7.09	0	3	12	104	40	56	6149	4961.58	97,693.28
manatee50-0.5	0.00	0.58	0.00	0.62	2.20	0.12	n/a	0	0.00	0	0	0	0	0	0	1	1.17	110,280.03
montgomery50-0.5	0.29	1.23	0.16	15.42	1.81	0.22	0.00	32	2.47	0	0	69	2	22	85	585	629.30	123,121.57
palmbeach50-0.5	5.34	0.68	3.91	7236.20	0.00	0.08	n/a	0	0.00	0	0	0	42	895	0	1	TL(3.91)	112,956.22
robeson50-0.5	0.24	0.79	0.00	1.38	12.13	0.09	n/a	0	0.00	0	0	0	2	2	0	1	1.78	111,067.43
sampson50-0.5	0.00	1.49	0.00	2.42	8.20	0.20	n/a	0	0.00	0	0	0	0	0	0	1	4.24	86,512.83
wayne50-0.5	0.00	0.56	0.00	0.56	7.79	0.03	n/a	0	0.00	0	0	0	0	0	0	1	0.84	96,641.59
$\tau/\gamma = 1.0$																		
ascension50-1.0	0.12	2.62	0.02	6.54	11.13	0.47	0.00	1	0.22	0	0	12	3	7	0	7	19.85	95,807.76
beaufort50-1.0	0.00	1.45	0.00	1.45	10.21	0.07	n/a	0	0.00	0	0	0	0	0	0	1	2.39	86,273.50
brazoria50-1.0	0.49	0.70	0.00	11.83	6.99	0.54	n/a	0	0.00	0	0	19	107	83	0	6	27.55	100,568.60
brevard50-1.0	1.54	1.72	0.00	4.17	13.51	0.08	n/a	0	0.00	14	25	0	1	6	0	1	5.12	82,036.51
brunswick50-1.0	1.13	1.85	0.83	3.29	3.96	0.07	0.00	13	0.40	1	14	50	0	6	0	312	396.56	102,240.55
harris50-1.0	0.00	0.57	0.00	0.62	5.29	0.05	n/a	0	0.00	0	0	0	0	0	0	1	1.08	134,687.78
jefferson50-1.0	0.79	1.22	0.57	6.18	5.18	0.04	0.00	30	0.58	0	0	22	1	127	2	539	618.79	118,528.07
lafourche50-1.0	1.66	1.63	0.08	3.41	4.44	0.03	0.00	3	0.05	22	14	19	0	4	0	16	34.88	111,824.81
lee50-1.0	0.01	0.62	0.01	0.63	3.44	0.05	0.00	1	0.02	0	0	0	0	0	0	3	1.67	106,951.12
manatee50-1.0	0.30	0.50	0.00	0.90	2.10	0.08	n/a	0	0.00	0	0	0	103	3	0	1	1.29	131,685.78
montgomery50-1.0	0.00	0.38	0.00	1.25	5.34	0.59	n/a	0	0.00	0	0	0	0	0	0	1	3.32	130,138.17
palmbeach50-1.0	0.00	0.49	0.00	0.50	8.11	0.03	n/a	0	0.00	0	0	0	0	0	0	1	0.69	122,111.74
robeson50-1.0	0.01	1.63	0.00	1.91	14.37	0.06	n/a	0	0.00	0	0	0	2	2	0	1	2.17	116,260.23
sampson50-1.0	0.20	2.18	0.00	14.39	10.17	0.83	n/a	0	0.00	0	0	0	111	95	0	1	17.18	95,502.94
wayne50-1.0	0.00	0.60	0.00	0.61	3.39	0.17	n/a	0	0.00	0	0	0	0	0	0	1	1.64	111,525.64
$\tau/\gamma = 1.5$																		
ascension50-1.5	0.14	2.58	0.11	6.93	4.26	0.51	0.00	3	0.21	0	0	28	5	23	110	376	403.57	108,706.83
beaufort50-1.5	0.21	1.60	0.20	1.89	10.60	0.07	0.00	4	0.06	0	0	17	5	23	26	303	289.33	99,132.58
brazoria50-1.5	0.32	0.66	0.12	3.32	3.42	0.21	n/a	0	0.00	0	0	57	0	4	0	10	22.17	110,338.42
brevard50-1.5	1.10	1.45	0.08	3.01	4.27	0.10	0.00	1	0.05	14	0	0	0	7	0	5	10.80	90,343.55
brunswick50-1.5	0.32	1.65	0.18	2.80	9.77	0.09	0.00	3	0.06	1	14	34	0	14	0	16	28.33	115,579.17
harris50-1.5	0.86	0.48	0.38	2.40	2.82	0.07	0.00	8	0.19	0	0	17	110	29	50	360	347.20	153,554.11
jefferson50-1.5	0.50	0.66	0.47	0.73	2.40	0.02	0.00	13	0.10	2	8	21	0	22	0	181	85.27	129,461.95
lafourche50-1.5	1.22	1.57	0.18	2.04	1.18	0.02	0.00	5	0.06	5	17	8	0	4	0	35	68.76	125,893.24
lee50-1.5	0.03	0.63	0.03	0.64	3.02	0.04	0.00	1	0.01	0	0	0	0	0	0	3	1.98	121,305.78
manatee50-1.5	0.00	0.43	0.00	0.48	2.93	0.05	n/a	0	0.00	0	0	0	0	0	0	1	0.78	150,830.17
montgomery50-1.5	0.03	0.71	0.00	3.35	7.47	0.11	n/a	0	0.00	0	0	20	0	8	0	1	5.11	146,055.32
palmbeach50-1.5	0.46	0.31	0.07	0.86	2.66	0.03	0.00	1	0.01	0	0	5	4	17	0	7	4.60	140,041.43
robeson50-1.5	1.63	0.53	0.90	17.46	9.50	0.13	0.00	163	43.07	0	0	5	7	162	188	7492	TL(0.20)	135,561.10
sampson50-1.5	0.03	0.54	0.02	2.16	3.57	0.35	n/a	0	0.00	0	0	0	0	2	0	3	5.97	106,667.92
wayne50-1.5	0.15	0.97	0.00	1.26	7.60	0.04	n/a	0	0.00	0	0	0	2	2	0	1	1.48	121,524.25

Table EC.14 Detailed Results on VTC Instance-Set-50 with Varying Cable-Trench Coefficients.

	Root				H_0		F_R			Total								
Instance	gap(LP)	CPU(LP)	gap(LPC)	CPU(LPC)	gap	CPU	gap	call	CPU	GCI	SRI(T)	SRI(R)	CLI	MWI	CFI	node	CPU(IP)	Z_{F_2}
ascension50-1	0.00	1.22	0.00	1.58	14.54	0.13	n/a	0	0.00	3	6	13	0	1	0	1	2.36	55,102.60
beaufort50-1	1.03	1.02	0.72	1.61	16.27	0.07	0.04	21	0.39	6	5	17	1	18	2	228	192.02	53,682.55
brazoria50-1	0.09	0.46	0.00	4.00	6.77	0.13	n/a	0	0.00	0	0	28	1	91	0	1	5.10	66,146.31
brevard50-1	2.22	1.03	0.05	5.38	20.92	0.04	0.04	1	0.02	26	40	0	0	11	0	3	22.88	55,978.13
brunswick50-1	1.76	1.49	1.10	3.08	11.05	0.05	0.00	126	2.50	3	12	49	0	7	0	3604	3134.73	67,006.38
harris50-1	0.00	0.39	0.00	0.43	10.28	0.03	n/a	0	0.00	0	0	0	0	0	0	1	0.71	72,953.64
jefferson50-1	0.90	0.88	0.38	1.15	15.54	0.05	n/a	0	0.00	5	8	6	0	6	0	3	3.28	70,897.59
lafourche50-1	2.27	3.34	0.07	12.29	24.21	0.06	0.00	2	0.05	28	34	0	1	1	0	5	44.80	66,895.00
lee50-1	0.68	0.75	0.67	1.21	7.83	0.07	0.00	62	48.17	0	25	34	4	47	10	1770	1564.21	67,195.51
manatee50-1	0.00	0.52	0.00	0.56	5.33	0.10	n/a	0	0.00	0	0	0	0	0	0	1	0.93	64,614.13
montgomery50-1	0.31	0.67	0.28	4.12	3.57	0.16	0.00	107	6.70	0	0	83	1	10	200	1143	752.48	83,106.17
palmbeach50-1	0.00	0.36	0.00	0.37	7.45	0.02	n/a	0	0.00	0	0	0	0	0	0	1	0.59	59,663.94
robeson50-1	0.23	1.62	0.00	2.84	17.12	0.10	n/a	0	0.00	0	0	0	102	1	0	1	3.14	91,998.28
sampson50-1	0.55	1.04	0.12	21.49	12.88	0.17	0.00	15	1.30	1	8	121	104	173	26	240	646.83	62,431.73
wayne50-1	0.00	0.52	0.00	0.52	15.14	0.03	n/a	0	0.00	0	0	0	0	0	0	1	0.73	66,106.15
ascension50-2	0.18	1.10	0.00	1.71	25.88	0.16	n/a	0	0.00	3	4	10	0	3	0	1	2.60	49,541.57
beaufort50-2	0.00	1.56	0.00	1.56	24.12	0.10	n/a	0	0.00	0	0	0	0	0	0	1	2.26	54,425.21
brazoria50-2	0.08	0.84	0.03	7.54	13.74	0.50	0.00	4	0.32	0	0	5	6	11	0	15	26.49	54,924.32
brevard50-2	1.76	1.94	0.10	3.47	26.91	0.07	0.53	6	0.21	27	125	0	3	7	0	173	2387.17	52,367.87
brunswick50-2	2.13	0.83	1.45	1.81	9.87	0.04	0.00	115	2.34	4	27	25	0	4	0	1098	1097.43	59,974.64
harris50-2	0.00	0.58	0.00	0.62	12.56	0.05	n/a	0	0.00	0	0	0	0	0	0	1	0.94	72,939.47
jefferson50-2	1.58	1.08	1.20	2.93	11.94	0.03	0.00	94	3.47	5	21	14	458	1244	272	3912	4214.80	73,168.17
lafourche50-2	2.31	0.82	0.62	12.52	12.39	0.03	0.00	50	0.60	32	66	20	0	10	2	508	1995.55	56,851.26
lee50-2	0.01	0.51	0.01	0.52	8.01	0.03	0.00	1	0.01	0	0	0	0	0	0	3	1.66	63,010.79
manatee50-2	0.00	0.35	0.00	0.39	4.86	0.05	n/a	0	0.00	0	0	0	0	0	0	1	0.71	68,279.66
montgomery50-2	0.04	1.33	0.04	4.29	10.26	0.12	0.00	1	0.07	0	0	16	2	11	0	34	38.00	89,033.66
palmbeach50-2	0.00	0.43	0.00	0.44	20.46	0.02	n/a	0	0.00	0	0	0	0	0	0	1	0.65	57,641.28
robeson50-2	0.00	1.01	0.00	1.47	23.16	0.06	n/a	0	0.00	0	2	0	0	0	0	1	1.72	80,973.47
sampson50-2	0.44	1.61	0.23	17.58	14.70	0.22	0.00	4	0.36	1	8	83	4	71	0	46	155.46	66,083.62
wayne50-2	0.00	0.81	0.00	0.82	7.71	0.02	n/a	0	0.00	0	0	0	0	0	0	1	1.02	74,292.12
ascension50-3	0.00	1.38	0.00	1.49	9.76	0.14	n/a	0	0.00	0	0	0	0	0	0	1	2.35	54,350.02
beaufort50-3	0.32	0.82	0.24	0.90	20.45	0.11	0.11	1	0.01	7	4	0	0	0	0	3	3.21	59,544.86
brazoria50-3	0.05	0.60	0.00	2.28	4.71	0.14	n/a	0	0.00	0	0	0	0	2	0	1	3.56	63,486.73
brevard50-3	2.49	0.98	0.12	3.67	7.59	0.03	0.00	1	0.01	27	62	6	0	20	0	5	96.36	56,081.82
brunswick50-3	1.42	0.94	0.53	1.56	17.52	0.09	0.00	41	0.78	5	29	54	0	5	10	482	850.61	62,687.00
harris50-3	0.00	0.36	0.00	0.40	10.19	0.04	n/a	0	0.00	0	0	0	0	0	0	1	0.71	71,799.79
jefferson50-3	0.10	0.83	0.00	0.97	15.76	0.04	n/a	0	0.00	5	10	12	0	0	0	1	1.35	61,443.53
lafourche50-3	0.84	0.98	0.34	2.26	6.48	0.04	0.00	6	0.06	27	35	0	0	1	0	31	115.15	60,233.55
lee50-3	0.14	0.74	0.04	1.05	9.73	0.04	n/a	0	0.00	9	3	0	0	0	0	3	2.35	58,687.54
manatee50-3	0.11	0.49	0.00	0.80	12.75	0.06	n/a	0	0.00	0	0	13	0	20	0	1	1.17	68,423.22
montgomery50-3	0.24	0.79	0.18	14.38	17.02	0.51	0.00	20	1.10	0	0	77	102	20	12	169	245.08	86,436.69
palmbeach50-3	0.00	0.27	0.00	0.28	8.64	0.03	n/a	0	0.00	0	0	0	0	0	0	1	0.52	61,945.55
robeson50-3	0.00	1.28	0.00	1.29	18.48	0.06	n/a	0	0.00	0	0	0	0	0	0	1	1.55	90,572.68
sampson50-3	1.02	0.84	0.68	9.27	7.93	0.11	0.10	307	22.40	1	2	58	0	28	0	6388	TL(0.10)	67,237.61
wayne50-3	0.00	0.43	0.00	0.43	20.76	0.02	n/a	0	0.00	0	0	0	0	0	0	1	0.62	68,298.56

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